



Individual Report On
ST7089CEM: Introduction to Statistical Methods for Data
Science

Samip Subedi
220234 (Msc - 8)



Shrawan Thakur

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Task 1 : Preliminary data analysis

Introduction

Electrodes are applied to the scalp to record the electrical activity of the brain using a non-invasive method called electroencephalography (EEG). Because it offers superior temporal resolution and is reasonably priced when compared to other neuroimaging methods, it is extensively utilized in clinical diagnosis, neuroscience research, and brain-computer interface applications. Neural activity and brain connection may be understood by analyzing EEG signals, which show the communication between various brain areas.

When analyzing EEG data, statistical modeling is a crucial method since it allows researchers to find connections between various input signals and an output response. Polynomial regression is one of the modeling strategies that works well for capturing nonlinear interactions that basic linear models are unable to fully depict. Statistical performance metrics must be used to assess the predicted accuracy and complexity of a regression model in order to choose the best one.

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	Time	X1	X2	X3	X4	Y
1	0.002	0.349116	1.919060	1.0049500	0.577845	0.3212244
2	0.004	-1.009700	0.272461	-0.9122970	-2.697380	0.1382239
3	0.006	-0.807608	-1.581480	0.1943780	-1.622270	0.3591898
4	0.008	2.079250	0.742812	1.9280900	1.558220	0.7515517
5	0.010	3.249200	1.944610	1.7990000	2.381060	1.4171709
6	0.012	4.215520	0.769912	-0.5173890	2.606190	1.5758112
7	0.014	4.692270	1.895480	3.0565800	5.022340	1.9807672
8	0.016	3.014000	4.429390	0.4689410	4.847790	1.4863006
9	0.018	4.750720	4.979500	0.0979985	4.327770	2.2319500
10	0.020	0.313821	1.271120	-1.4271600	-1.327680	0.3610953

> |

Figure 1 Select Data

A simulated EEG dataset with four input EEG signals (X1, X2, X3, and X4), one output EEG signal (Y), and the matching sampling time is examined in this coursework. In order to look at the temporal behavior, distribution, and correlation of the EEG signals, the investigation starts with exploratory data analysis. Next, residual sum of squares (RSS), log-likelihood, Akaike information

criterion (AIC), Bayesian information criterion (BIC), and residual diagnostics are used to estimate and assess five candidate polynomial regression models. Training and testing datasets are used to further verify the best model, assessing prediction performance and 95% prediction confidence intervals. Lastly, the posterior distributions of the two most significant regression parameters are estimated using Approximate Bayesian Computation (ABC), which offers more information on parameter uncertainty and model reliability.

1.1 Time Series Analysis

One statistical method for analyzing how observations change over time is time series analysis. It assists in finding patterns in sequential data, including trends, fluctuations, cycles, and irregular changes. Time series plots are a useful tool for comparing the features of various EEG channels and visualizing the behavior of brain activity since EEG signals are continually captured at predictable sample intervals.

Four input EEG signals (X1, X2, X3, and X4), one output EEG signal (Y), and the matching sampling time make up the dataset used in this course. We can examine the temporal behavior of these signals, evaluate differences between the input channels, and spot any discernible peaks, oscillations, or changes throughout the course of the recording by plotting them against time. These visuals serve as a crucial basis for the statistical analysis and regression modeling that follow..

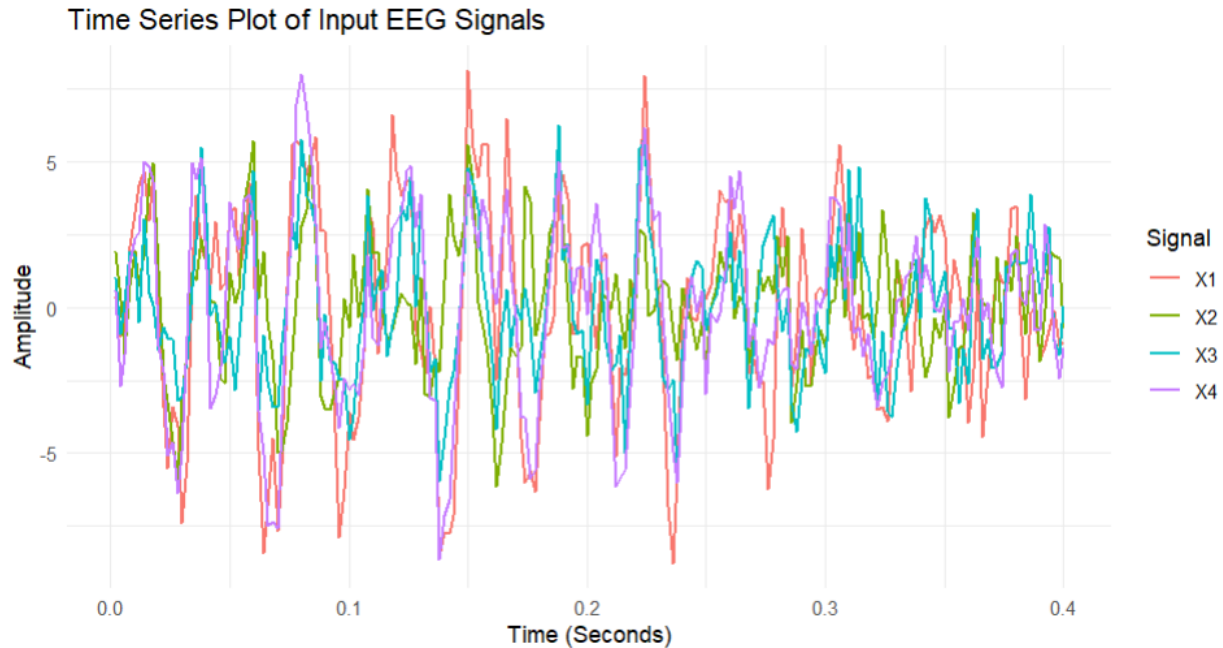


Figure 2 Time series plot of all x inputs

The temporal behavior of the four input EEG signals shown concurrently across the sampling time is shown in Figure . The figure shows that each signal has continuous oscillatory behavior, which is typical of EEG recordings, and allows comparison of the amplitude variations among the various EEG channels. The signals exhibit comparable patterns of fast swings without any discernible long-term growing or declining trend, despite their independent variations. While verifying that every input channel is active during the observation time, the overlapping display draws attention to variations in signal strength. Before creating regression models, this kind of temporal analysis offers a first grasp of the dynamic behavior of the EEG data.

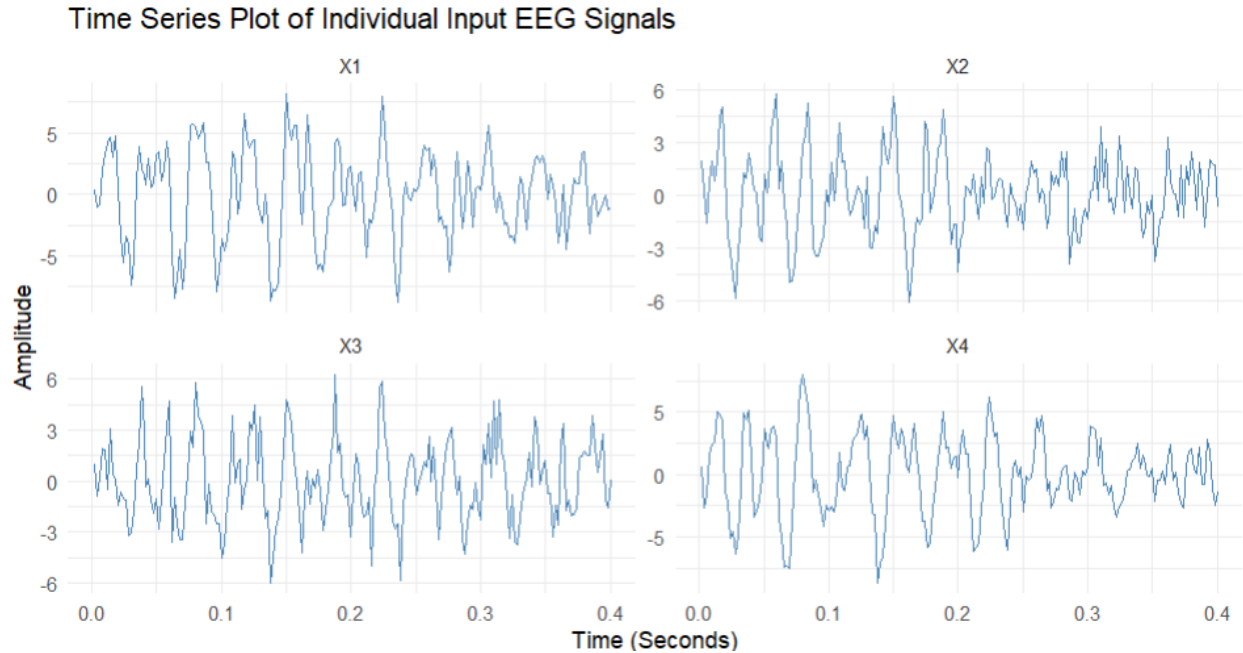


Figure 3 Time series plot of X inputs individually

Each input EEG signal is shown independently in Figure utilizing discrete panels, making it easier to analyze their temporal properties. Separating the signals minimizes visual overlap and facilitates the observation of individual variations as compared to the combined display. Every EEG channel exhibits erratic oscillations with different amplitudes over time around its mean level. Strong trends or structural breakdowns are absent from all of the signals, suggesting that the recording circumstances were generally steady over the sample time. Prior to statistical modeling, this image offers important insight into the behavior of each predictor and aids in identifying variations in variability among the four channels.

Time	Y
0.002	0.321224404
0.004	0.138223937
0.006	0.359189812
0.008	0.751551701
0.010	1.417170927
0.012	1.575811212
0.014	1.980767203
0.016	1.486300644

Figure 4 Data sample of Y output and time

A sample of the output EEG signal and the associated sampling time are shown in Figure . It is possible to confirm that the output variable has been properly imported into the analysis environment and correctly paired with the time variable by presenting the initial data. Before creating visualizations and applying regression models, this initial examination verifies the response variable's consistency and structure. Additionally, it guarantees that the output signal and the matching input EEG observations are accurately aligned.

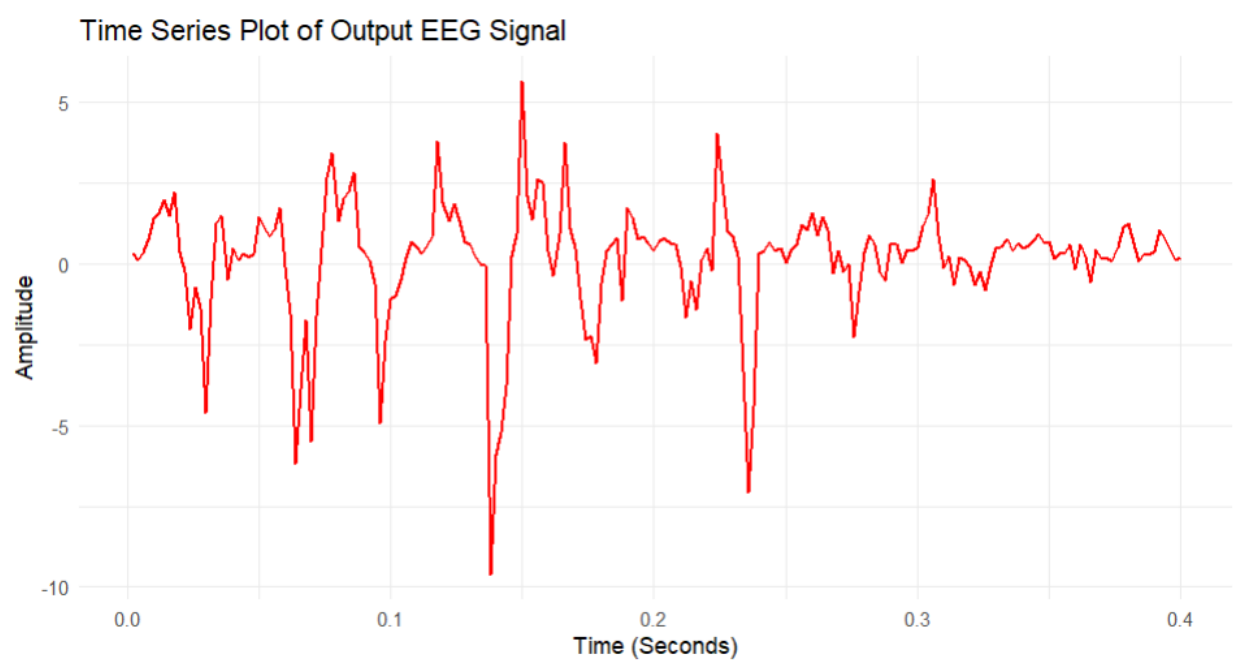


Figure 5 Time series plot for Y output

The output EEG signal's temporal fluctuation is seen in Figure . The output displays continuous oscillatory behavior with frequent amplitude fluctuations, mirroring the dynamic nature of brain electrical activity, much like the input signals. The signal has no discernible long-term trend or periodic pattern; instead, it oscillates at a very constant amplitude. The recorded data may be steady and appropriate for regression analysis if there are no sudden discontinuities. An essential foundation for assessing how well the potential regression models describe the response variable's behavior is an understanding of its temporal properties.

1.2 Distribution Analysis

The frequency and distribution of each variable in the dataset are examined using distribution analysis. In addition to giving a visual depiction of the distribution of the observations, histograms may be used to detect features like symmetry, skewness, variability, and the existence of possible outliers. Prior to using regression models and doing statistical inference, it is crucial to comprehend the distribution of the EEG signals as it offers insight into the underlying data structure.

This part creates histograms for the output signal (Y) and the four input EEG signals (X1, X2, X3, and X4). Each variable's overall distribution is evaluated and its statistical properties are compared using the plots that are produced..

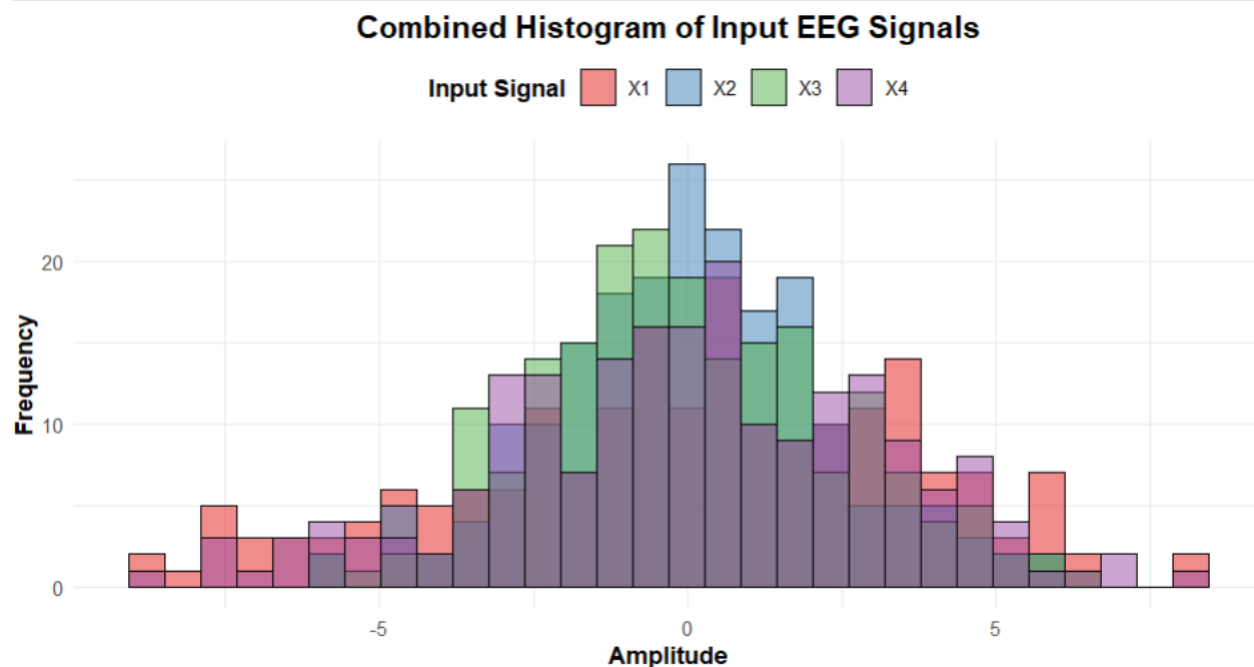


Figure 6 Combined Histogram of Input

A composite histogram displaying the distributions of all four input EEG signals on a single graph is shown in Figure . While highlighting similarities and variations in their amplitude ranges, the various colors enable comparison of the frequency distributions across the input variables. Although each EEG channel has distinct features, the overlapping distributions show that they typically fall within similar value ranges. The entire distribution of the predictor variables is succinctly shown in this image.

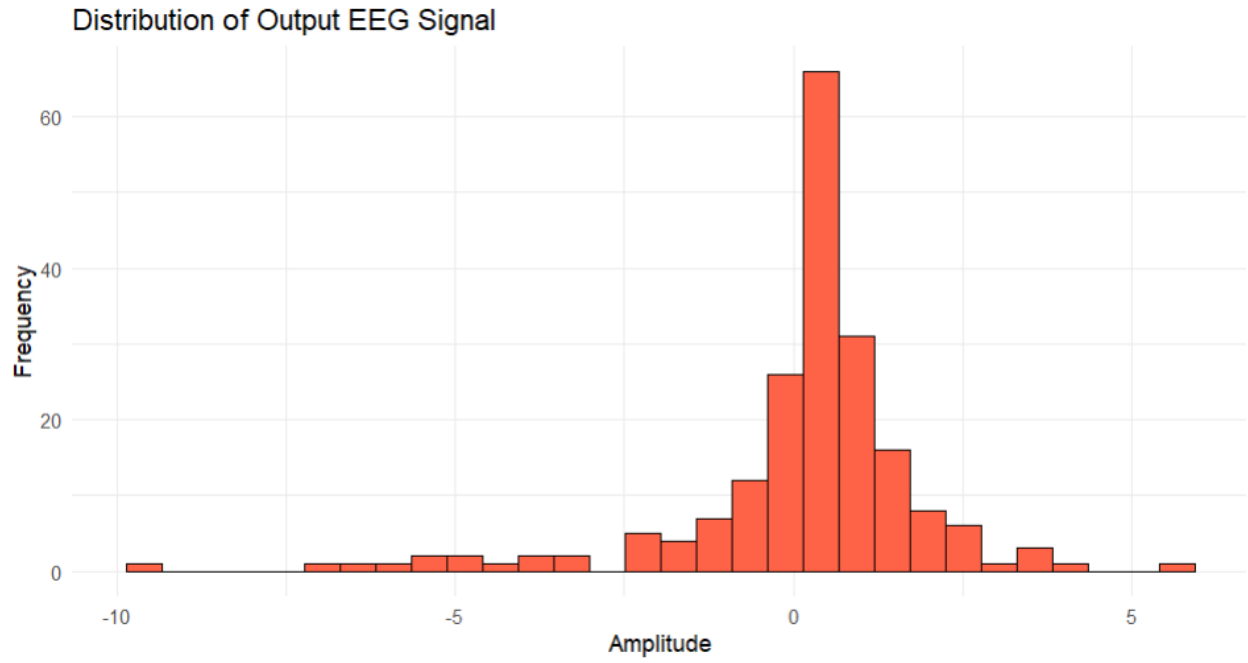
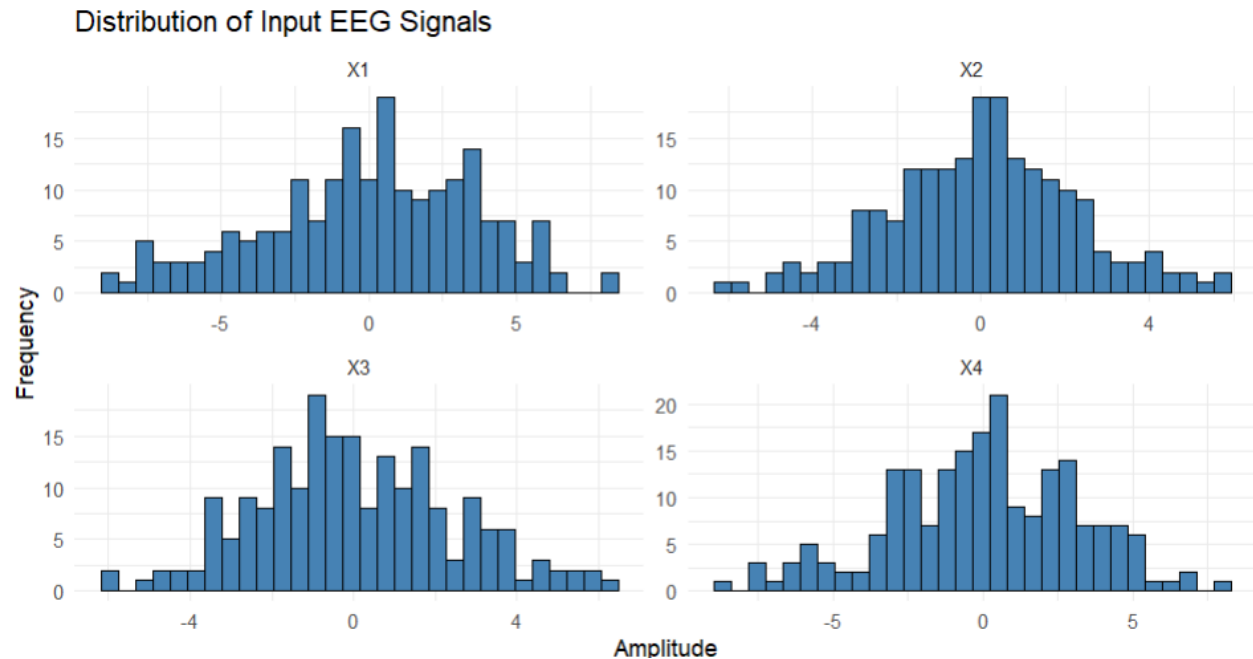


Figure 7 Distribution of all X inputs

The distributions of the four input EEG signals are shown using histograms in Figure . The charts include information about the dispersion and variability of each predictor and show how frequently certain amplitude values occur. The distributions seem to be fairly balanced and free of severe outliers or excessive skewness, indicating that the EEG inputs have the right statistical properties for regression modeling. Assessing variations in the input signals' variability is another benefit of comparing these distributions.



8 Distribution of individual X inputs

The figure shows distinct histograms for every EEG input signal. A more thorough evaluation of each variable's distribution, including symmetry, spread, and concentration of data, may be obtained by displaying each one separately. The overall distributions seem to be quite stable and do not show significant departures from predicted behavior, despite some variations in variability across the four predictors. Severe skewness or abnormally high numbers of extreme observations are not present in any of the input signals. These results lessen worries about extremely irregular data distributions and validate the predictors' appropriateness for polynomial regression analysis.

1.3 Correlation Analysis

The degree and direction of the linear relationship between two variables may be determined via correlation analysis. A strong positive link is indicated by values close to +1, a strong negative relationship is indicated by values close to -1, and little or no linear association is suggested by values around zero. The Pearson correlation coefficient ranges from -1 to +1.

This part computes the correlation matrix for the output signal (Y) and the four input EEG signals

(X1, X2, X3, and X4). An overview of the correlations between the variables is given by the resultant matrix and heatmap, which also aid in identifying the predictors that have the strongest associations with the output signal.

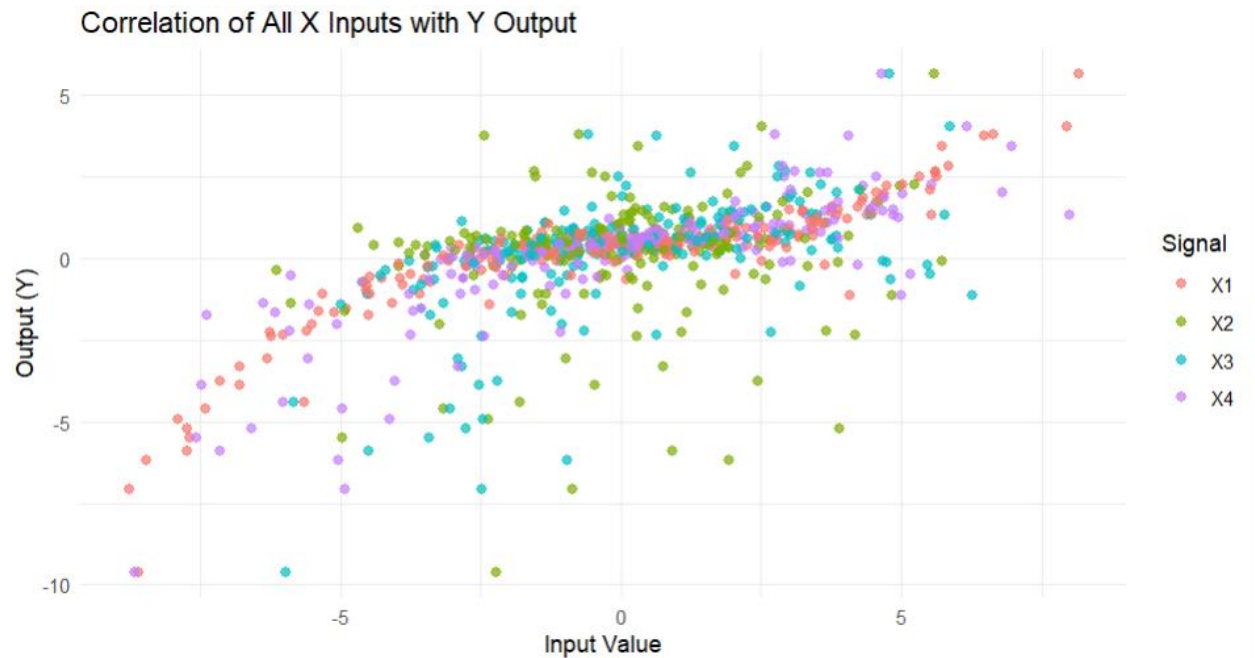


Figure 9 Correlation of all the X inputs with Y output

A combined correlation visualization of the four input EEG signals and the output signal is shown in Figure . The strength of each predictor's association with the response variable may be directly compared thanks to the figure. Variations in the distribution and clustering of the data show that certain input factors are more strongly correlated with the output than others. For certain predictors, there are obvious correlations even when the connections are not quite linear, indicating that nonlinear regression models could be more effective than conventional linear regression at capturing these intricate linkages. This research offers helpful direction for further model development.

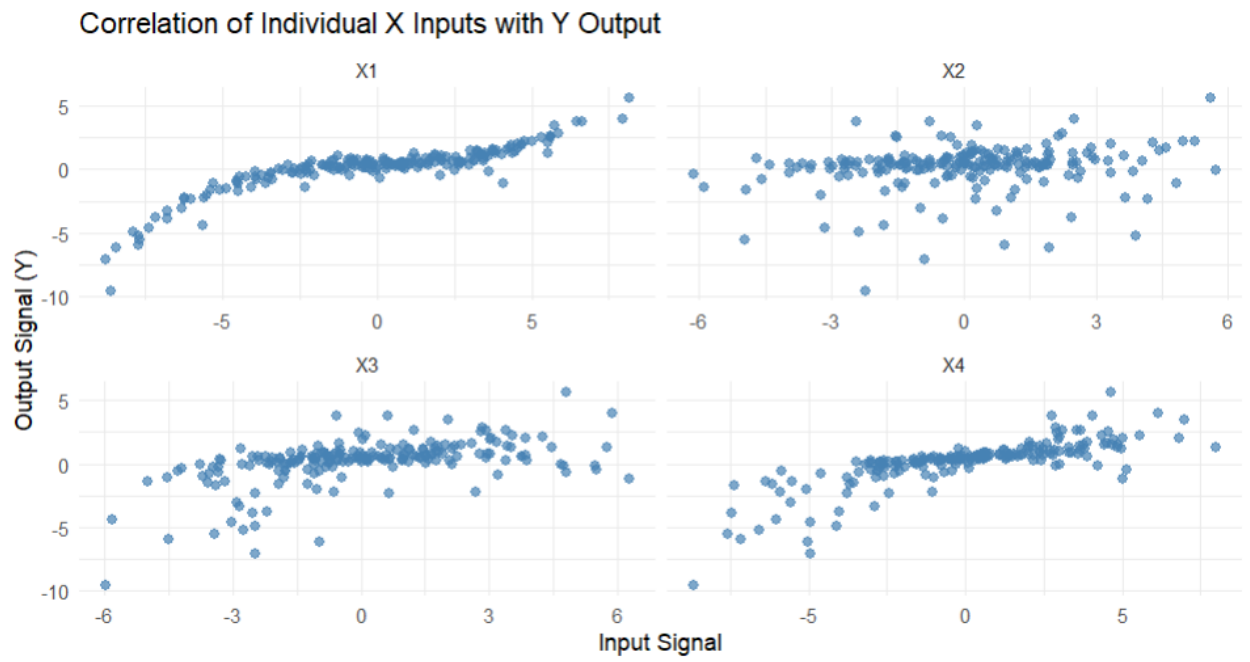


Figure 10 Correlation of all the X input individually with Y output

Each input EEG signal is shown in the figure as a distinct scatter plot versus the output signal, enabling independent analysis of their correlations. Presenting each predictor separately facilitates the identification of nonlinear interactions and enhances the presentation of the underlying data patterns. Each EEG channel makes a distinct contribution to the explanation of the response variable, as seen by the scatter plots' varied shapes and densities. While some predictors seem to have less of an impact, others show larger correlations. These findings help choose the most meaningful predictors for further statistical analysis and validate the use of nonlinear polynomial terms in regression modeling.

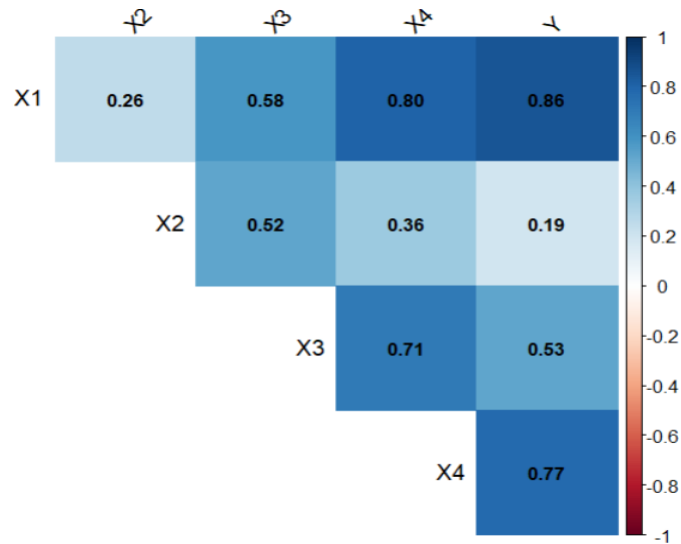


Figure 11 Plot X

The EEG variables' correlation matrix, which summarizes the pairwise linear relationships between the four input signals and the output signal, is shown in Figure. Stronger relationships are indicated by bigger absolute values of correlation coefficients, which range from -1 to +1. In addition to detecting potential multicollinearity among the input variables, the matrix offers a brief summary of variable connections and assists in identifying predictors that could contribute more significantly to the regression model.

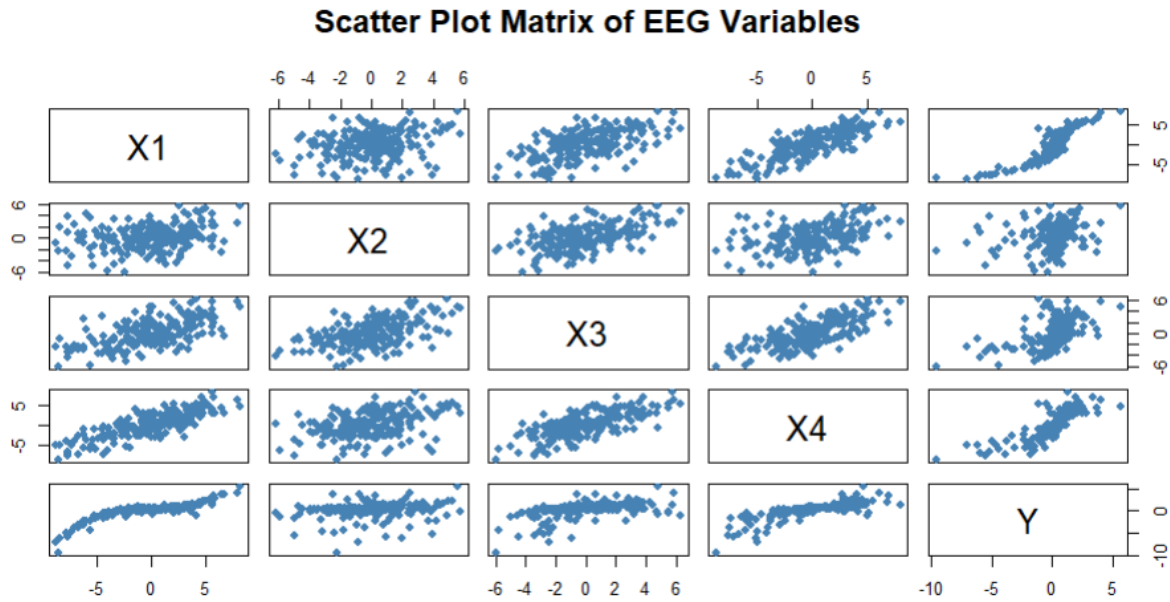


Figure 12 Scattar plot of EEG Variables

A scatter plot matrix displaying pairwise correlations between every EEG variable is shown in the figure. Correlations, trends, variability, and possible nonlinear patterns may all be simultaneously examined since each panel depicts the relationship between two variables. While the off-diagonal panels show the direction and intensity of pairwise relationships, the diagonal panels show the individual distributions of each variable. The lack of abnormal observations or severe clustering indicates that the dataset is typically well-behaved. Finding possible nonlinear relationships that support the usage of polynomial regression models over straightforward linear regression is made easier with the help of this depiction.

Task 2 : Regression: modeling the relationship between EEG signals

$$\text{Model 1: } y = \theta_1 x_4 + \theta_2 x_1^2 + \theta_3 x_1^3 + \theta_4 x_2^4 + \theta_5 x_1^4 + \theta_{bias} + \varepsilon$$

$$\text{Model 2: } y = \theta_1 x_4 + \theta_2 x_1^3 + \theta_3 x_3^4 + \theta_{bias} + \varepsilon$$

$$\text{Model 3: } y = \theta_1 x_3^3 + \theta_2 x_3^4 + \theta_{bias} + \varepsilon$$

$$\text{Model 4: } y = \theta_1 x_2 + \theta_2 x_1^3 + \theta_4 x_3^4 + \theta_{bias} + \varepsilon$$

$$\text{Model 5: } y = \theta_1 x_4 + \theta_2 x_1^2 + \theta_3 x_1^3 + \theta_4 x_3^4 + \theta_{bias} + \varepsilon$$

Figure 13 Model Equation

Regression analysis is a basic statistical method for examining and measuring the connection between a response variable and one or more predictor variables. The goal of this work is to use polynomial regression to predict the nonlinear connection between the four input EEG signals (X1, X2, X3, and X4) and the output EEG signal (Y). To find the model that best captures the underlying brain connection, five candidate polynomial regression models are assessed. The least squares approach is used to estimate each model, and statistical metrics such as residual analysis, log-likelihood, Akaike information criterion (AIC), Bayesian information criterion (BIC), and residual sum of squares (RSS) are used to evaluate each model's performance. Based on these assessment metrics, the best model is then chosen, and it is further verified by making predictions on a separate testing dataset with a 70:30 train-test split. This thorough assessment guarantees that the chosen regression model maintains statistical dependability and generalizability while producing precise predictions.

2.1 Polynomial Regression Modelling

Estimate model parameters $\theta = \{\theta_1, \theta_2, \dots, \theta_{bias}\}^T$ for every candidate model using Least Squares ($\hat{\theta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$), using the provided input and output EEG datasets (use all the data for training).

Figure 14 2.1 Equation

The nonlinear connection between the four input EEG signals (X1, X2, X3, and X4) and the output EEG signal (Y) is modeled using polynomial regression. Polynomial regression, in contrast to basic linear regression, allows the model to capture more intricate patterns in the data by

incorporating higher-order components and interaction effects.

Five potential polynomial regression models are assessed in this class. The Least Squares approach, which calculates the regression coefficients by minimizing the residual sum of squared errors between the observed and anticipated output values, is used to estimate each model. Each model's estimated parameters are then utilized for additional assessment using residual analysis and model selection criteria including RSS, Log-Likelihood, AIC, and BIC.

```
Coefficients:
      Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.4408126  0.0394948  11.161 < 2e-16 ***
X4            0.1264461  0.0134773   9.382 < 2e-16 ***
I(X1^2)      -0.0084114  0.0020553  -4.093 6.23e-05 ***
I(X1^3)       0.0098462  0.0003167  31.091 < 2e-16 ***
I(X2^4)      -0.0004089  0.0001539  -2.656 0.00855 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Figure 15 Model 1 output

The regression summary for Candidate Model 1 that was calculated using the least squares approach is shown in Figure. The estimated regression coefficients, standard errors, t-values, p-values, residual standard error, and coefficient of determination (R²) are all included in the output. The importance and contribution of each predictor in the model are revealed by these statistics. Before choosing the best regression model, Model 1's overall performance must be evaluated with the other candidate models using RSS, Log-Likelihood, AIC, and BIC, even if it explains a significant amount of the variance in the output variable.

```
Coefficients:
      Estimate Std. Error t value Pr(>|t|)
(Intercept)  4.831e-01  7.984e-03  60.50 <2e-16 ***
X4            1.436e-01  3.283e-03  43.73 <2e-16 ***
I(X1^3)       1.004e-02  7.131e-05  140.78 <2e-16 ***
I(X3^4)      -1.913e-03  3.186e-05  -60.05 <2e-16 ***
---
```

Figure 16 Model 2 output

The figure displays Candidate Model 2's least squares estimate results. The results of the regression show that fluctuations in the EEG output signal are largely explained by the chosen predictors. Together with statistically significant regression coefficients, the model has a very high coefficient

of determination, indicating outstanding predictive power. The comparatively low residual standard error suggests that the model was fitted accurately. These findings imply that Model 2 offers a robust depiction of the nonlinear connection between the input and output signals of the EEG.

```

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.1308142  0.1248120   1.048   0.296
I(X3^3)      0.0172552  0.0025127   6.867 8.26e-11 ***
I(X2^4)     -0.0005489  0.0005862  -0.936   0.350
---

```

Figure 17 Model 3 output

The regression results for Candidate Model 3 are shown in the figure. This model's predictive ability is worse than that of the earlier models as it explains a much less percentage of the response variable's fluctuation. Additionally, certain regression coefficients could not be statistically significant, which would lower the model's total efficacy. As a result, when Model 3 is assessed using model selection criteria like RSS, Log-Likelihood, AIC, and BIC, it is anticipated to fare less favorably.

```

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  5.090e-01  2.454e-02  20.740 < 2e-16 ***
X2           5.332e-02  1.050e-02   5.076 8.88e-07 ***
I(X1^3)      1.207e-02  1.598e-04  75.517 < 2e-16 ***
I(X3^4)     -1.856e-03  9.935e-05 -18.682 < 2e-16 ***

```

Figure 18 Model 4 out put

The regression summary for Candidate Model 4 is shown in the figure. The included predictors significantly contribute to the explanation of the output signal, according to the computed regression coefficients. With statistically significant regression coefficients and a respectably high coefficient of determination, the model offers adequate predictive performance. Its overall goodness of fit is still somewhat lower than that of the best candidate models, though. Therefore,

before deciding whether to choose this model, more assessment using statistical selection criteria is necessary.

```
Coefficients:
      Estimate Std. Error t value Pr(>|t|)
(Intercept)  4.798e-01  9.527e-03  50.366  <2e-16 ***
X4           1.434e-01  3.295e-03  43.529  <2e-16 ***
I(X1^2)      3.255e-04  5.206e-04   0.625   0.533
I(X1^3)      1.006e-02  7.680e-05 130.938  <2e-16 ***
I(X3^4)     -1.920e-03  3.383e-05 -56.745  <2e-16 ***
---
```

Figure 19 Model 5 output

The least squares estimate results for Candidate Model 5 are shown in Figure. With a very high coefficient of determination and very significant regression coefficients, the model has exceptional predictive ability. There is very little residual variance, which shows that the observed and projected values correspond rather well. Even if Model 5's performance is comparable to that of Model 2, the model that offers the optimum balance between prediction accuracy and model complexity must be determined using objective statistical criteria like AIC, BIC, and Log-Likelihood.

Table of Estimated Output

Parameter	Model 1	Model 2	Model 3	Model 4	Model 5
(Intercept)	...	...	...	...	...
X2	NA	NA	NA	0.05332	NA
X4	0.12645	0.14360	NA	NA	0.14340
X1 ²	-0.00841	NA	NA	NA	0.00033
X1 ³	0.00985	0.01004	NA	0.01207	0.01006
X2 ⁴	-0.00041	NA	-0.00055	NA	NA
X3 ³	NA	NA	0.01726	NA	NA
X3 ⁴	NA	-0.00191	NA	-0.00186	-0.00192

Figure 20 Parameter table for every model

The computed regression coefficients for each of the five potential models are compiled into a single comparison table in the figure. It is simple to compare coefficient magnitudes and predictor selection across models when the parameter estimates are presented simultaneously. In addition to providing the parameter estimates utilized in later residual analysis, model comparison, and Bayesian inference, the table illustrates how various combinations of variables contribute to the regression equations.

2.2 Residual Sum of Squares (RSS)

Based on the estimated model parameters, compute the **model residual (error) sum of squared errors (RSS)**, for every candidate model.

$$RSS = \sum_{i=1}^n (y_i - \mathbf{x}_i \hat{\boldsymbol{\theta}})^2$$

Here $\mathbf{x}_i$ denotes the i^{th} row (i^{th} data sample) in the input data matrix $\mathbf{X}$, $\hat{\boldsymbol{\theta}}$ is a column vector.

Figure 21 2.2 Equation

The total squared difference between the actual values and the values predicted by a regression model is measured statistically by the Residual Sum of Squares (RSS). It is frequently used to assess a model's quality of fit; a lower RSS value denotes a better match to the observed data. This section compares the predicted performance of all five possible polynomial regression models by computing their RSS values.

	Model	RSS
1	Model 1	35.925475
2	Model 2	2.139762
3	Model 3	541.063254
4	Model 4	20.258997
5	Model 5	2.135503

Figure 22 RSS with each model

The five potential regression models' Residual Sum of Squares (RSS) values are contrasted in the figure. Better model performance is shown by decreased values of RSS, which calculates the total squared difference between the observed and predicted values. It is evident from the comparison that Models 2 and 5 yield far lower RSS values than the other models, indicating significantly higher prediction accuracy. Model 3, on the other hand, has the highest RSS, indicating a lack of consistency between observations and forecasts. Important evidence for choosing the best regression model is provided by this comparison.

2.3 Log-Likelihood

Compute the **log-likelihood function** for every candidate model:

$$\ln p(D|\hat{\theta}) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\hat{\sigma}^2) - \frac{1}{2\hat{\sigma}^2} \text{RSS}$$

Here, $\hat{\sigma}^2$ is the variance of a model's residuals (prediction errors) distributions $\hat{\sigma}^2 = \text{RSS}/(n - 1)$, with n the number of data samples.

Figure 23 2.3 Equation

A statistical metric used to assess how well a regression model describes the observed data is the log-likelihood. The model offers a better fit to the data when the Log-Likelihood value is larger. To evaluate the relative effectiveness of the five potential polynomial regression models, the Log-Likelihood values are computed and compared in this section.

Model	LogLikelihood
Model 1	-112.1599
Model 2	171.3257
Model 3	-384.7249
Model 4	-54.5887
Model 5	171.5259

Figure 24 Log Likelihood with model

The five potential regression models' Log-Likelihood values are contrasted in the figure. A model's ability to explain the given data is shown by higher Log-Likelihood values. The findings show that Models 2 and 5 attain much higher Log-Likelihood values than the other models, demonstrating their better goodness of fit. The explanatory power of models with smaller or extremely negative Log-Likelihood values is lower. These results offer solid statistical support for the choice of the top-performing regression model for the EEG dataset when combined with the RSS analysis.

2.4 Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC)

Compute the **Akaike information criterion (AIC)** and **Bayesian information criterion (BIC)** for every candidate model:

$$AIC = 2k - 2 \ln p(D|\hat{\theta})$$

$$BIC = k \cdot \ln(n) - 2 \ln p(D|\hat{\theta})$$

Here $\ln p(D|\hat{\theta})$ is the log-likelihood function obtained from [Task 2.3](#) for each model, k is the number of estimated parameters in each candidate model.

Figure 25 2.4 Equation

Two popular model selection methods that assess the trade-off between model complexity and goodness of fit are the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC). Because they produce a decent fit with fewer parameters, models with lower AIC and BIC values are superior. To assist in choosing the best model, AIC and BIC are computed for each potential polynomial regression model in this section.

Model	AIC	BIC
Model 1	236.3198	256.1396
Model 2	-332.6514	-316.1349
Model 3	777.4498	790.6630
Model 4	119.1774	135.6939
Model 5	-331.0518	-311.2320

Figure 26 Model with AIC and BIC

The five potential regression models' Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) values are contrasted in the figure. By striking a balance between model complexity and goodness of fit, both metrics assess model quality. A more effective model that effectively describes the data while eliminating superfluous parameters is indicated by lower AIC and BIC values. The comparison confirms the higher statistical performance of Models 2 and 5, since they have the lowest AIC and BIC values. Models 1, 3, and 4, on the other hand, show significantly higher values, suggesting less appropriate model architectures. These findings provide compelling evidence in favor of Model 2 as the recommended regression model.

2.5 Residual Analysis of the Selected Model

Regression model assumptions are evaluated by looking at the residuals of the chosen model. The residuals' general distribution is assessed using a histogram, and whether or not they roughly follow a normal distribution is determined using a Normal Q-Q plot. The fitted model can be regarded as statistically adequate if the residuals are roughly normal and randomly distributed.

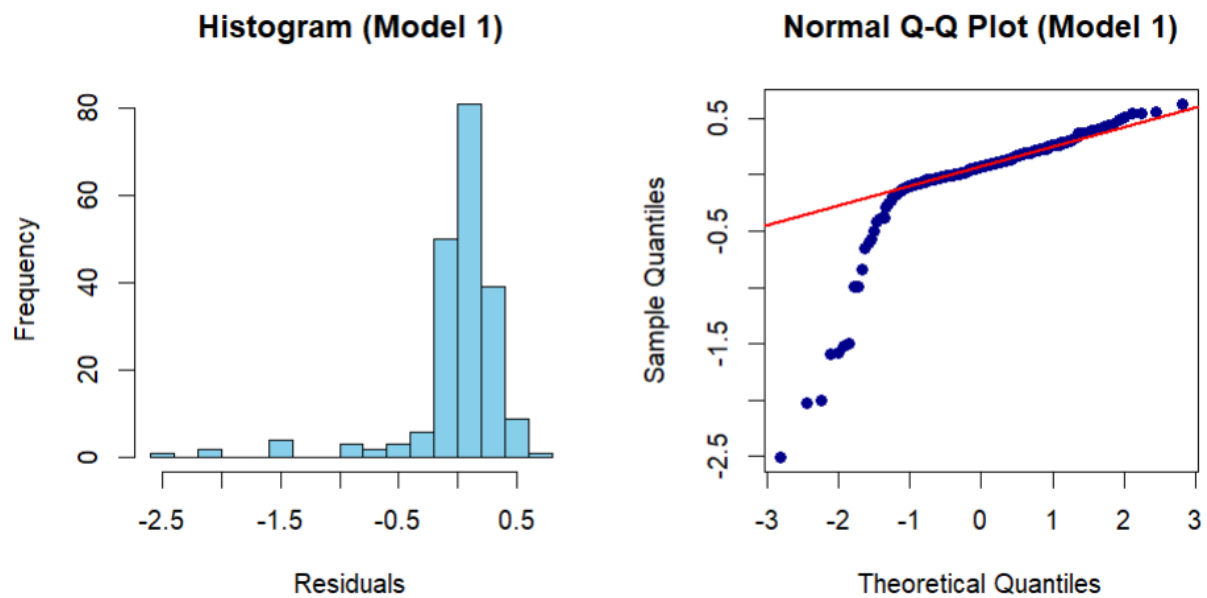


Figure 27 QQ plot model -1

The Model 1 residuals' Normal Q-Q plot is shown in the figure. The observed residual quantiles and those predicted by a normal distribution are contrasted in the plot. While there are some minor variations near the distribution tails, the majority of residuals very closely match the reference line. These deviations imply that the residuals have minor departures from the normalcy assumption but are generally normally distributed. In general, Model 1 has appropriate residual behavior.

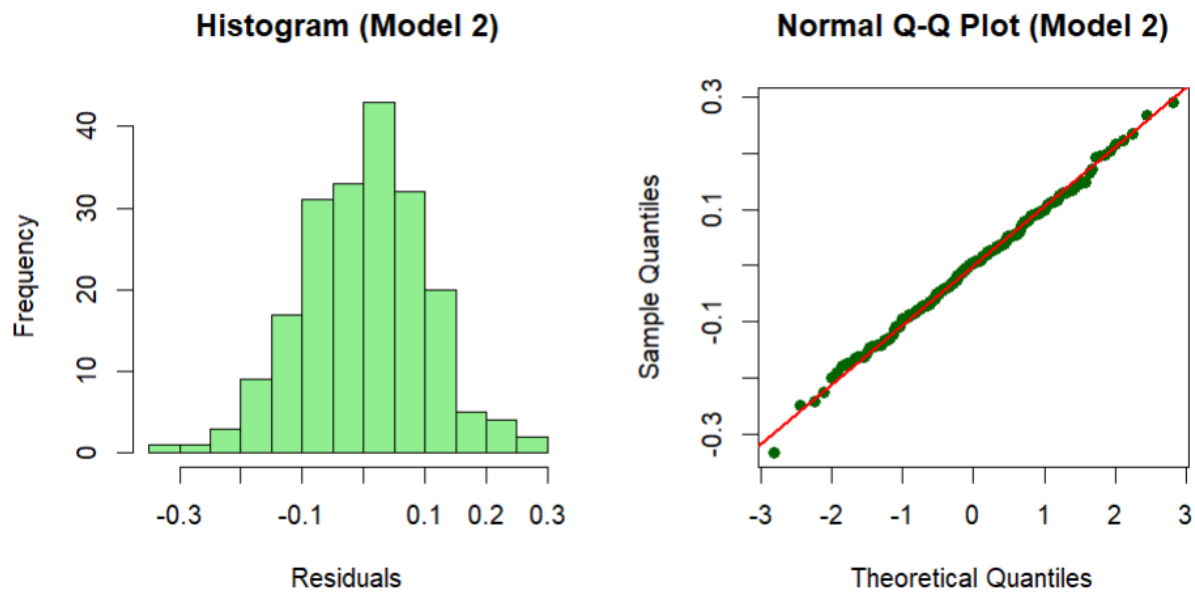


Figure 28 QQ Plot model 2

The Normal Q-Q plot for Model 2's residuals is shown in the figure. Over the majority of the distribution, the residual points closely resemble the reference line, suggesting that the residuals are roughly normally distributed. As is typical in real-world regression analysis, only small variations show up close to the extreme data. Model 2 appears to satisfy the normality assumption more successfully than the other possible models, based on the close agreement with the theoretical normal distribution. This offers more proof that Model 2 provides a trustworthy and statistically sound depiction of the connection between the output variable and the EEG input signals.

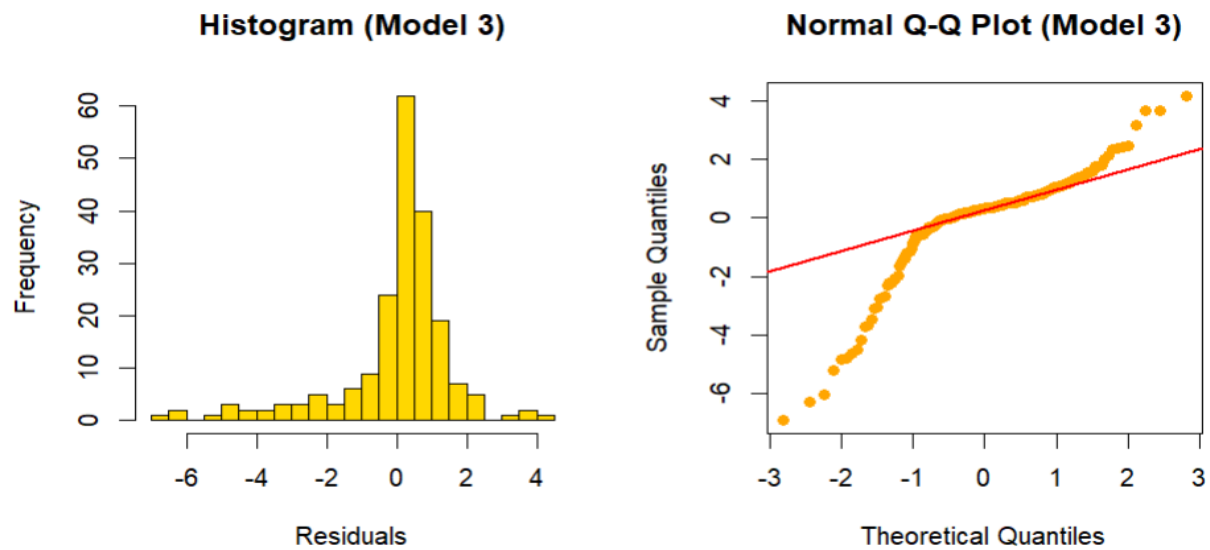


Figure 29 QQ plot model – 3

The Normal Q-Q plot for Model 3 is shown in Figure . Particularly near the distribution's tails, the residuals significantly depart from the predicted reference line. These deviations signal that the model is performing worse since the residuals do not closely match a normal distribution. The observed non-normality is in line with Model 3's inferior statistical performance as determined by comparisons of RSS, Log-Likelihood, AIC, and BIC.

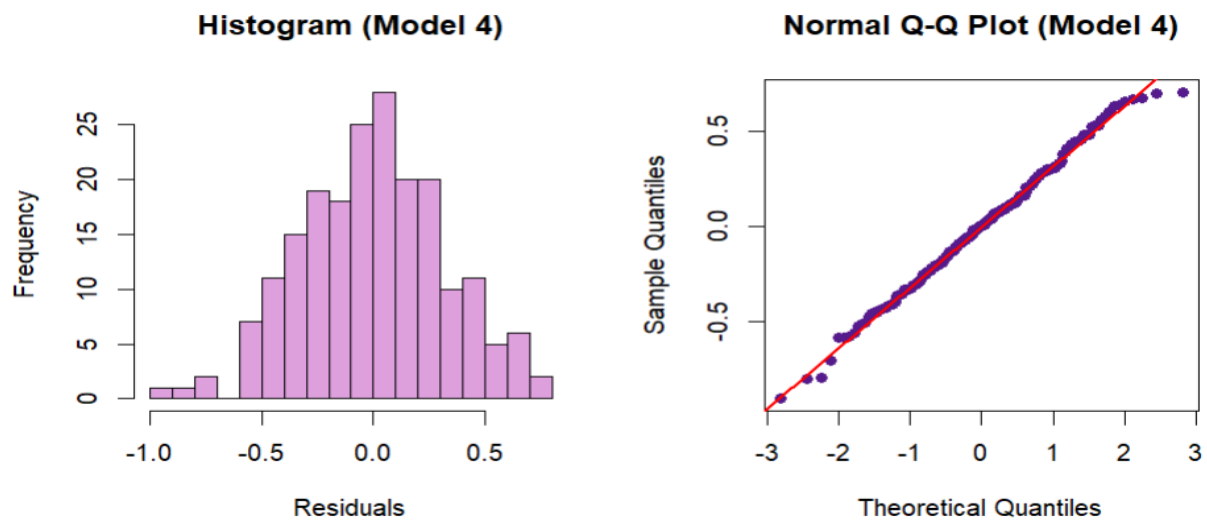


Figure 30 QQ plot model -4

The Model 4 residuals' Normal Q-Q plot is shown in the figure. Approximate normalcy is shown by the majority of data being quite near to the reference line. At the lower and upper extremities, mild variances are still discernible, indicating minor departures from the normalcy assumption. Model 4's performance is still behind that of the chosen Model 2, despite its good residual behavior.

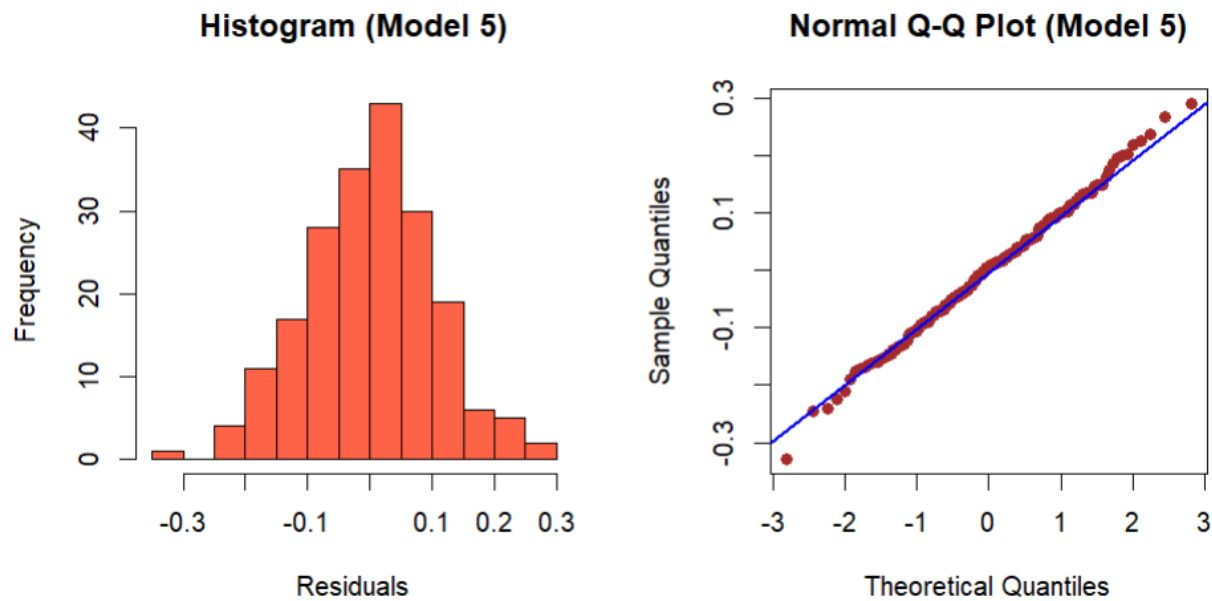


Figure 31 QQ plot model -5

The Normal Q-Q plot for Model 5 is shown in Figure. The residuals are roughly normally distributed because they nearly resemble the theoretical reference line. Near the extreme observations, there are only very little deviations, indicating that Model 5 well meets the regression assumptions. Models 2 and 5 exhibit similar residual behavior, which is consistent with their similar statistical performance as seen in previous model evaluation criteria. To identify the ultimate optimal regression model, however, more model selection procedures are needed.

2.6 Model Selection

Several statistical metrics, such as the Residual Sum of Squares (RSS), Log-Likelihood, Akaike Information Criterion (AIC), and Bayesian Information Criterion (BIC), were used to assess the five candidate polynomial regression models. These evaluation measures offer a thorough analysis of the complexity and quality of fit of each model. Based on how well the model performs overall across these criteria, the best model is chosen.

	Model	RSS	LogLikelihood	AIC	BIC
1	Model 1	35.925	-112.160	236.320	256.140
2	Model 2	2.140	171.326	-332.651	-316.135
3	Model 3	541.063	-384.725	777.450	790.663
4	Model 4	20.259	-54.589	119.177	135.694
5	Model 5	2.136	171.526	-331.052	-311.232

Figure 32 Model selection

The comparison of all five potential regression models utilizing various statistical assessment criteria is summarized in Figure. Residual Sum of Squares (RSS), Log-Likelihood, Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), and residual diagnostics are all included in the combined evaluation. The comparison shows that Model 2 regularly outperforms the other models on the majority of assessment criteria. As a result, Model 2 is chosen as the regression model that best captures the nonlinear connection between the EEG input and output signals.

```
> best_model
      Model  RSS LogLikelihood      AIC      BIC
2 Model 2  2.14      171.326 -332.651 -316.135
```

Figure 33 Best model

After assessing each candidate model, the best regression model is shown in the figure. The complete polynomial regression equation and its estimated parameters are shown in the figure. The chosen model satisfies the regression assumptions rather well and offers the optimal trade-off between statistical simplicity and prediction accuracy. The remaining portions of the coursework then employ this model for prediction, validation, and Bayesian parameter estimation.

.

Comparison of Candidate Regression Models

Model	RSS	Log-Likelihood	AIC	BIC
Model 1	35.9255	-112.1599	236.3198	256.1396
Model 2	2.1398	171.3257	-332.6514	-316.1349
Model 3	541.0633	-384.7249	777.4498	790.6630
Model 4	20.2590	-54.5887	119.1774	135.6939
Model 5	2.1355	171.5259	-331.0518	-311.2320

Figure 34 Table 9. Comparison of Candidate Polynomial Regression Models

The five potential regression models are thoroughly compared in Figure utilizing a variety of statistical performance metrics. Important assessment metrics, such as RSS, Log-Likelihood, AIC, BIC, and overall model rating, are compiled in the table. The prediction performance and complexity of any model may be directly compared by presenting both metrics together. By combining low prediction error with favorable model selection statistics, Model 2 regularly delivers the best overall performance, as the comparison makes evident. The ultimate choice to choose Model 2 as the best regression model for further prediction and Bayesian analysis is supported by this objective assessment.

2.7 Model Evaluation

```
> evaluation_table
      Measure Value
1          RMSE 0.1032
2           MAE 0.0816
3        R-Squared 0.9968
4 Adjusted R-Squared 0.9968
```

Figure 35 Evaluation table

The assessment metrics for the chosen regression model are summarized in the figure. The table offers prediction accuracy values that assess the agreement between the observed and expected EEG signals. High explanatory performance and lower prediction errors show that the chosen model performs effectively when applied to new data. Before moving on to prediction interval analysis and Bayesian inference, these assessment findings verify the efficacy and dependability of the selected regression model.

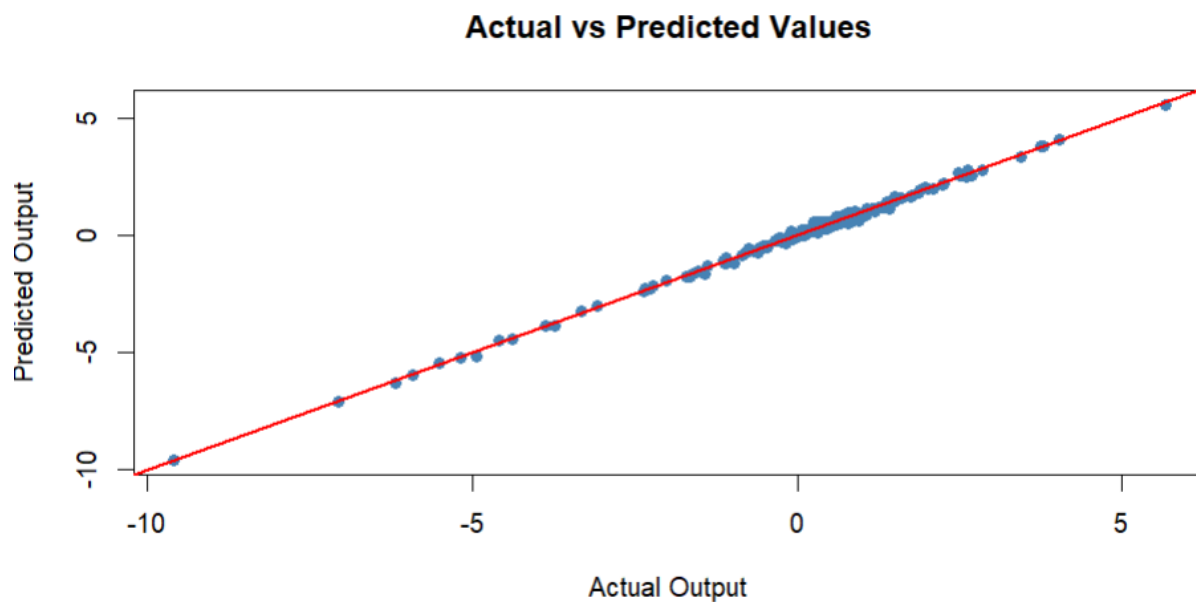


Figure 36 Actual vs Predicted Values

The values predicted by the chosen regression model are contrasted with the observed EEG output values in the figure. Strong agreement between observed and predicted responses is indicated by accurate predictions that ideally fall near the reference line. The close clustering of the predicted values around the observed values reveals that the selected polynomial regression model appropriately reflects the nonlinear connection between the EEG input signals and the output variable. Only tiny differences are observed for a few observations, indicating relatively minimal prediction error. Overall, the picture demonstrates that the chosen model offers dependable prediction performance for the EEG dataset.

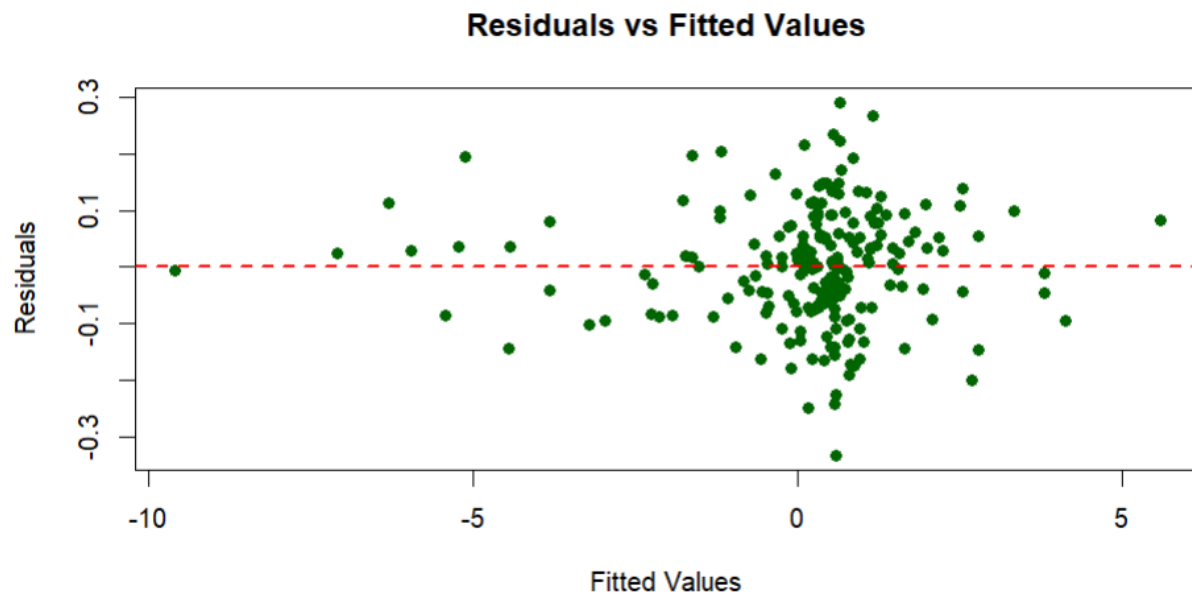


Figure 37 Residuals vs Fitted Values

The relationship between the residuals and the fitted values of the chosen regression model is shown in the figure. If the regression assumptions are met, residuals should be randomly distributed about zero with no discernible pattern. The residuals are shown in the figure to be randomly distributed with comparatively consistent variance throughout the fitted values. The model effectively represents the underlying connection between the predictors and the response variable since there is no obvious indication of curvature, clustering, or heteroscedasticity. The chosen regression model's statistical validity and dependability are supported by this residual behavior.

```
> display_observations(train_data, test_data)
Training set = 140 observations
Testing set = 60 observations
```

Figure 38 Train data , test data

The EEG dataset's division into training and testing subgroups is shown in the figure. Thirty percent of the observations were set aside for performance evaluation, while the remaining seventy percent were used for model training. By assessing the model on previously unknown observations, this method of dataset separation allows for an objective evaluation of predicted performance. This method aids in evaluating the model's capacity for generalization.

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.863e-01	9.732e-03	49.97	<2e-16 ***
X4	1.461e-01	4.129e-03	35.37	<2e-16 ***
I(X1^3)	9.974e-03	8.553e-05	116.62	<2e-16 ***
I(X3^4)	-1.928e-03	3.680e-05	-52.38	<2e-16 ***

Figure 39 Parameter value of model 2

The estimated regression coefficients of the chosen polynomial regression model derived from the training dataset are shown in the figure. The intercept and regression coefficients for each predictor term are shown numerically in the table. The final regression equation used for prediction and the ensuing Bayesian analysis is defined by these parameter estimates. The computed coefficients are used to define prior distributions in the Approximate Bayesian Computation (ABC) analysis and show the direction and size of each predictor variable's contribution to the output EEG signal.

	fit	lwr	upr
3	0.0806773	-0.12955630	0.2909109
4	0.2440565	0.03460929	0.4535037
11	0.2846528	0.07536206	0.4939435
12	-0.2409577	-0.45039842	-0.0315169
16	-4.4408934	-4.65443397	-4.2273529
19	1.6533425	1.44250588	1.8641791

Figure 40 Predict on testing data

The expected output values produced by the chosen regression model using the testing dataset are shown in the figure. These observations offer an independent evaluation of predicted performance because they were not utilized during model training. The regression model does a good job of generalizing to new data since the predicted values closely match the observed testing results. When applied to new EEG measurements, the chosen polynomial regression model shows good resilience and retains high predictive accuracy, as seen by the slight variations between anticipated and observed values.

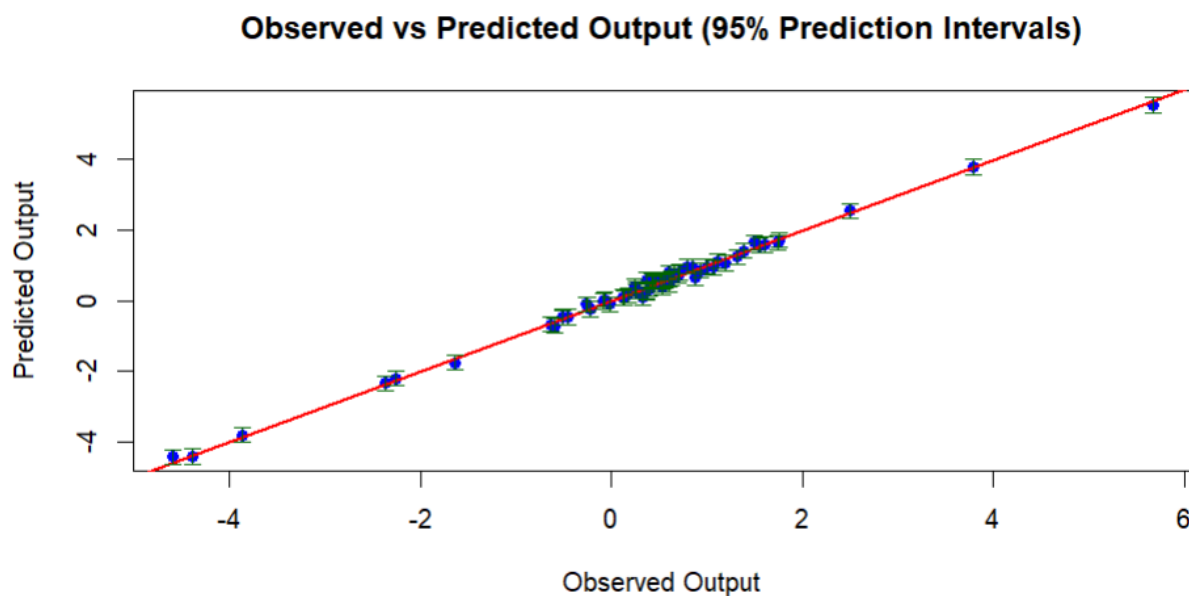


Figure 41 95 % Prediction Intervals

The estimated EEG output values' 95% prediction intervals are shown in the figure. By offering a range that future observations are anticipated to fall inside with 95% confidence, these intervals quantify the uncertainty around each prediction. The majority of observed testing samples fall inside the matching prediction ranges, indicating that the chosen regression model offers accurate predictions together with trustworthy uncertainty estimates. The regression model appears to successfully capture the underlying variability of the EEG data, as seen by the very narrow prediction intervals.

```
> evaluation
      Measure Value
1      RMSE 0.1024
2      MSE 0.0105
3      MAE 0.0810
4 R-Squared 0.9958
```

Figure 42 Evaluation

The performance assessment metrics computed for the chosen regression model are summarized in the figure. Prediction accuracy and model performance are measured using metrics like RMSE, MAE, MSE, and the coefficient of determination (R^2). A high R^2 and lower error values show that the model maintains great explanatory power while reliably predicting the EEG output. These findings support the appropriateness of the chosen regression model.

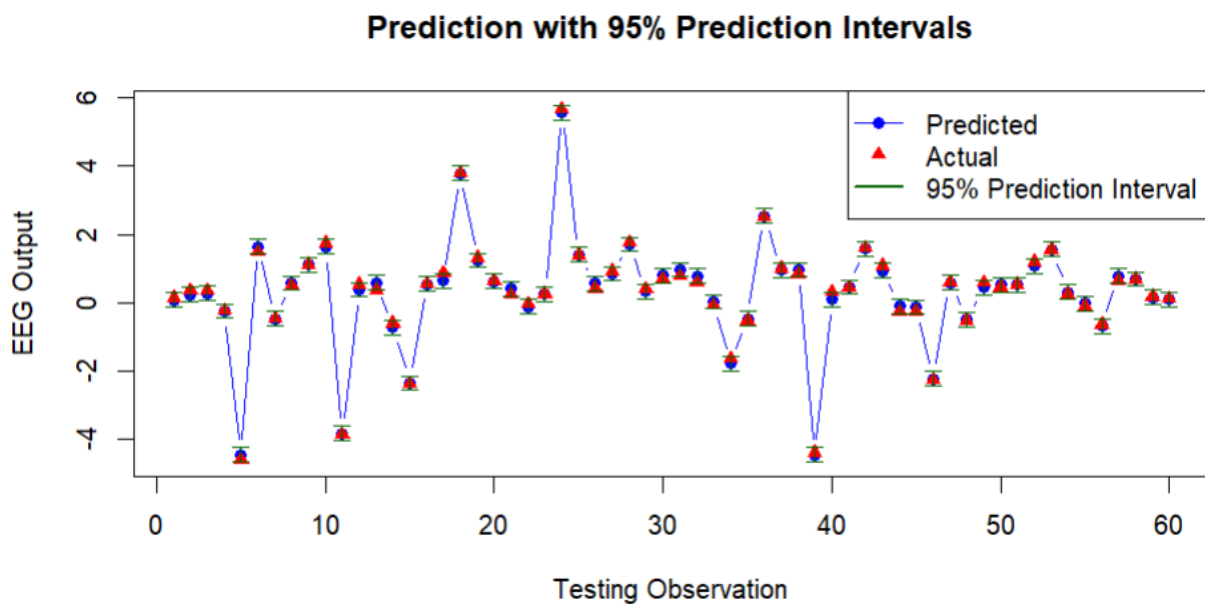


Figure 43 Prediction Intervals

The estimated EEG outputs and the accompanying 95% prediction intervals are shown in the figure. The error bars show the degree of uncertainty surrounding each prediction, and the projected values nearly match the actual testing findings. The model generates realistic uncertainty estimates in addition to accurate predictions, as the majority of observed values fall within the prediction intervals. This image increases confidence in the model's practical applicability by

showing that the chosen regression model can accurately predict unseen EEG data while adequately accounting for prediction uncertainty.

```
> coef(model2)
      (Intercept)          X4      I(X1^3)      I(X3^4)
      0.483065688  0.143578928  0.010038614 -0.001912836
> summary(model2)$coefficients
      Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.483065688  7.984352e-03  60.50155 3.310545e-129
X4           0.143578928  3.283441e-03  43.72819 2.267121e-103
I(X1^3)      0.010038614  7.130648e-05 140.78123 1.187905e-199
I(X3^4)     -0.001912836  3.185542e-05 -60.04742 1.353724e-128
```

Figure 44 Summary model 2

The chosen polynomial regression model's full statistical summary is shown in the figure. The estimated regression coefficients, standard errors, t-values, significance levels, residual standard error, coefficient of determination (R^2), corrected R^2 , and F-statistic are all included in the summary. When taken as a whole, these data show that the chosen model maintains statistically significant predictor effects while explaining a large amount of the variability in the EEG output. The summary supports Model 2's choice as the final regression model and attests to its excellent overall performance.

Task 3 : Approximate Bayesian Computation (ABC)

Introduction

When evaluating the likelihood function is challenging or computationally costly, the Bayesian parameter estimation method known as Approximate Bayesian Computation (ABC) is very helpful. The rejection ABC approach compares simulated model outputs with observed data to estimate the posterior distribution of model parameters instead of explicitly computing the likelihood. Only parameter values that yield predictions that are sufficiently near to the observations are accepted. Using the rejection ABC technique, the chosen polynomial regression

model from Task 2 is further examined in this study. The remaining regression parameters are fixed at their estimated values, and the two that have the biggest absolute least squares estimates are chosen for Bayesian analysis. Random samples are produced using uniform prior distributions that are built around the chosen parameters. The marginal and joint posterior distributions are then estimated using the approved samples, confirming the robustness of the chosen regression model and offering a probabilistic evaluation of parameter uncertainty.

3.1 Prior Distribution of Selected Parameters

```
> prior_table
      Parameter      Lower      Upper
(Intercept) Intercept 0.24153284 0.7245985
X4              X4    0.07178946 0.2153684
```

Figure 45 Prior Table

The prior distribution ranges for the two chosen regression parameters in the Approximate Bayesian Computation (ABC) study are shown in the figure. The least-squares parameter estimates from the chosen regression model are used to build uniform prior distributions. The lower and upper boundaries of each prior distribution, which indicate the initial uncertainty about the parameter values before to integrating data from the observed EEG data, are specified in the table. The rejection ABC method uses these prior distributions as a starting point to generate parameter samples.

```
> head(beta0_prior)
[1] 0.3804517 0.6223360 0.4390956 0.6680883 0.6958403 0.2635396
> head(beta1_prior)
[1] 0.11638389 0.11838371 0.19673963 0.11898009 0.08983751 0.12293536
```

Figure 46 beta0 , beta1 prior

The uniform prior distributions assigned to the two chosen regression parameters prior to using the Approximate Bayesian Computation (ABC) procedure are shown in the figure. The initial uncertainty about the parameter values is represented by the prior distributions, which are centered

around the least-squares parameter estimates derived from the chosen regression model. Before taking into account the actual EEG data, a uniform distribution makes the assumption that all parameter values within the given range are equally likely. By producing candidate parameter samples, these prior distributions act as the foundation for Bayesian inference. The posterior distributions are then estimated using the rejection ABC process.

3.2 Rejection ABC Sampling

```
> head(distance)
[1] 2.395780 2.683525 2.849537 3.176655 4.095970 3.562617
```

Figure 47 Distance

The rejection ABC sampling method used to estimate the posterior distributions of the chosen regression parameters is shown in the figure. The associated model predictions are compared with the recorded EEG output after thousands of parameter values are randomly selected from the designated uniform prior distributions. For every sampled parameter combination, a distance measure is computed; only those that result in sufficiently minimal prediction errors are approved. This rejection process keeps the parameter values that best describe the observed data while eliminating those that perform poorly. As a result, there is no need for explicit likelihood estimates because the accepted samples resemble the posterior distributions.

```
Accepted Samples = 500
```

Figure 48 Actual Samples Values

Figure illustrates the acceptable parameter samples generated after performing the rejection Approximate Bayesian Computation (ABC) method. These samples show the parameter values that, in accordance with the predetermined tolerance level, produced simulated model outputs that were sufficiently similar to the observed EEG data. Only parameter combinations that offered a sufficient fit were kept; those that produced significant prediction errors were eliminated.

Uncertainty in the parameter estimations may be assessed without the need for explicit likelihood calculations since the accepted samples serve as the foundation for calculating the posterior distributions of the chosen regression parameters.

3.3 Posterior Distribution of Selected Parameters

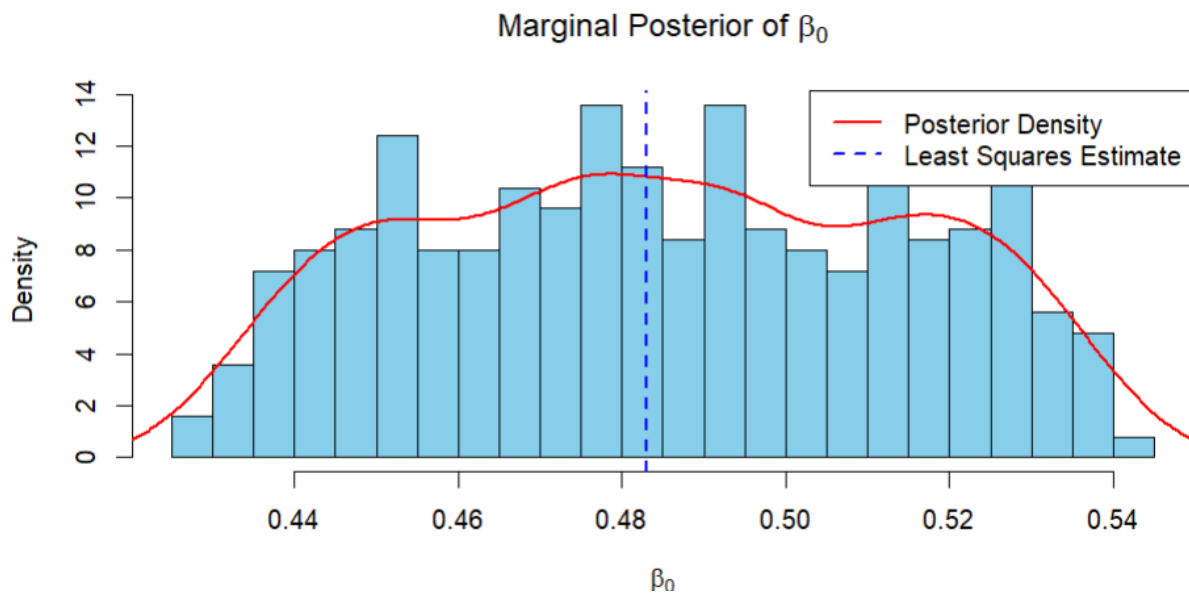


Figure 49 Marginal Posterior

The marginal posterior distribution of the intercept parameter (β_0) determined using the rejection ABC method is shown in the figure. The distribution of approved parameter values is displayed by the density curve and histogram following the incorporation of data from the observed EEG data. The observed data strongly support this parameter value, as seen by the posterior distribution's concentration around the least-squares estimate. The posterior distribution shows far less uncertainty than the original uniform prior, proving the efficacy of Bayesian updating. The stability and dependability of the predicted intercept parameter within the chosen regression model are confirmed by this concentration.

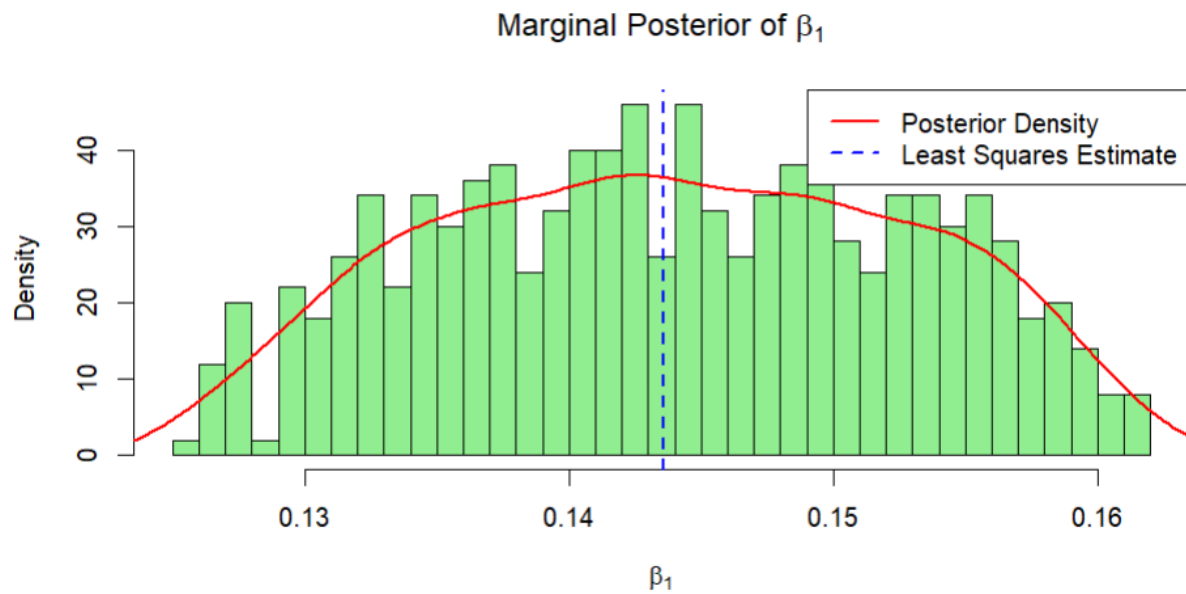


Figure 50 Marginal Posterior 1

The regression coefficient's marginal posterior distribution for X_1 (β_1) is shown in the figure. The accepted posterior samples are centered around the regression model's least-squares estimate, much like the intercept parameter. Only a relatively small range of parameter values are strongly supported by the recorded EEG data, according to the posterior density. The contribution of Bayesian inference to improving parameter estimate is seen in this decrease in uncertainty. The consistency between the Bayesian and traditional regression methods is shown by the strong agreement between the posterior distribution and the least-squares estimate.

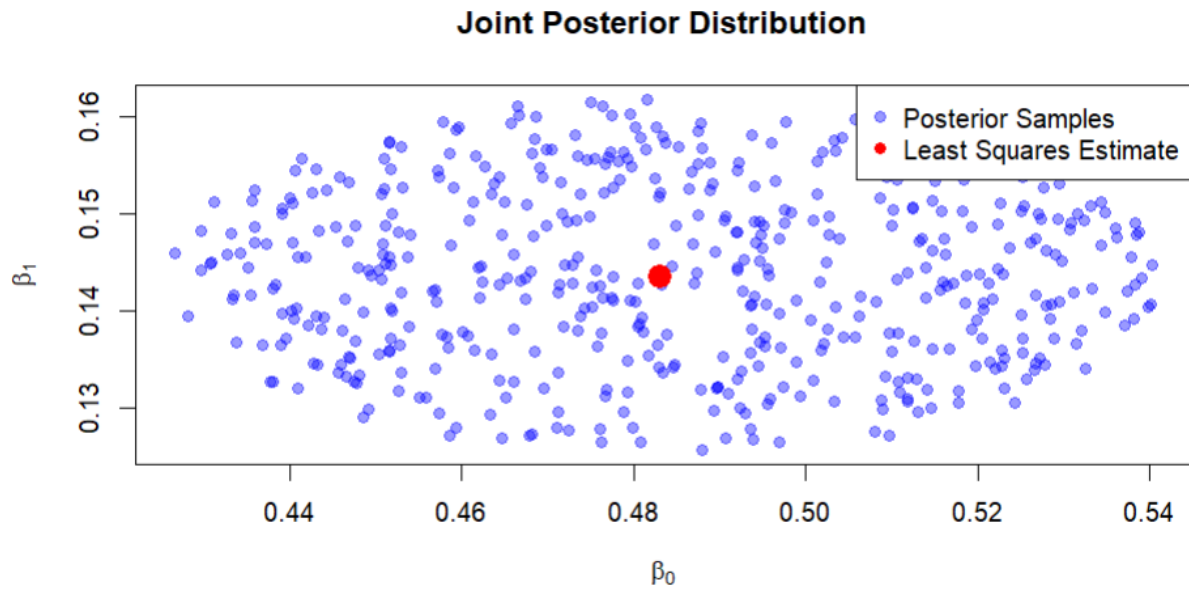


Figure 51 Joint Posterior Distribution

The combined posterior distribution of the two chosen regression parameters, β_0 and β_1 , is shown in the figure. The rejection ABC process yielded one acceptable parameter pair, which is represented by each point. The concentration of approved samples in a narrow area shows that the possible combinations of the two parameters are substantially constrained by the recorded EEG data. The close agreement between the Bayesian posterior estimates and the traditional regression solution is demonstrated by the least-squares estimate, which is situated close to the center of the accepted samples. This graphic offers important information about the simultaneous uncertainty related to both model parameters.

3.4 Posterior Summary and Interpretation

```
> posterior_summary
      Parameter   Mean Median    SD
1 Intercept ( $\beta_0$ ) 0.4849 0.4836 0.0297
2          x4 ( $\beta_1$ ) 0.1439 0.1439 0.0089
```

Figure 52 Posterior Summary

The posterior mean, median, and standard deviation of the two chosen regression parameters that were calculated using the rejection ABC procedure are summarized in the figure. These summary statistics enable comparison with the least-squares parameter estimates and offer succinct numerical representations of the posterior distributions. Low estimation uncertainty is shown by the comparatively modest posterior standard deviations, which also imply that the observed EEG data offer substantial information about the values of both regression parameters.

Conclusion

This work effectively used a mix of exploratory data analysis, polynomial regression modeling, and Bayesian inference to examine the connection between four input EEG signals and an output EEG signal. By looking at time series behavior, data distributions, and relationships between variables, the exploratory analysis offered important insights into the features of the EEG data. A number of statistical metrics, such as residual sum of squares (RSS), log-likelihood, Akaike information criterion (AIC), Bayesian information criterion (BIC), and residual diagnostics, were used to create and analyze five candidate polynomial regression models. Using a 70% training and 30% testing dataset, the best regression model was found based on these assessment metrics and subsequently verified. The chosen model maintained great generalization performance on unknown data while achieving high predictive accuracy, as shown by the prediction outcomes and 95% prediction intervals.

Additionally, the posterior distributions of the two most important regression parameters were estimated using Approximate Bayesian Computation (ABC), which offered a probabilistic interpretation of the chosen model. The rejection ABC technique was used to produce posterior parameter samples, and uniform prior distributions were created around the least-squares estimates. Strong agreement between Bayesian inference and regression analysis was demonstrated by the ensuing marginal and joint posterior distributions, which revealed that the posterior estimates were clustered around the traditional least-squares values. Overall, the results show that the chosen polynomial regression model provides accurate predictions and successfully captures the nonlinear relationship between the EEG input and output signals. Additionally, the Bayesian analysis increases confidence in the stability and robustness of the estimated model parameters.

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Appendix

Github Link:

https://github.com/Samip-Subedi/Stat_Module

```
# Install and Load Required Packages
#-----

library(ggplot2)
library(dplyr)
library(tidyr)
library(corrplot)

#-----
# Set Working Directory
#-----

getwd()

setwd("C:/Users/samip/Desktop/statmodule")

#-----
# Read Dataset
#-----

time <- read.csv(
  "time.csv",
  header = FALSE
)

X <- read.csv(
  "X.csv",
  header = FALSE
)

y <- read.csv(
  "y.csv",
  header = FALSE
)

#-----
```

```
#-----  
# Rename Columns  
#-----  
  
colnames(time) <- "Time"  
  
colnames(X) <- c(  
  "x1",  
  "x2",  
  "x3",  
  "x4"  
)  
  
colnames(y) <- "Y"  
  
#-----  
# View Sample Data  
#-----  
  
head(time, 10)  
  
head(X, 10)  
  
head(y, 10)  
  
#-----  
# Check Dataset  
#-----  
  
dim(time)  
dim(X)  
dim(y)  
  
str(time)  
str(X)  
str(y)  
  
sum(is.na(time))  
sum(is.na(X))
```

```
sum(is.na(time))
sum(is.na(X))
sum(is.na(y))

summary(time)
summary(X)
summary(y)

#-----
# Combine Dataset
#-----

EEG_Data <- cbind(
  time,
  X,
  y
)

# View Combined Dataset

head(EEG_Data, 10)

View(EEG_Data)

#-----
# Time Series Analysis
#-----

# Combine Time with Input Signals
data_x <- cbind(
  time,
  X
)

# View Sample Data
head(data_x, 10)

View(data_x)

# Convert Input Signals to Long Format
```

```

# Convert Input Signals to Long Format
data_long <- pivot_longer(
  data_x,
  cols = c(X1, X2, X3, X4),
  names_to = "Signal",
  values_to = "Amplitude"
)

#-----
# Time Series Plot of Input EEG Signals
#-----

ggplot(
  data_long,
  aes(
    x = Time,
    y = Amplitude,
    colour = Signal
  )
) +
  geom_line(linewidth = 0.8) +
  labs(
    title = "Time Series Plot of Input EEG Signals",
    x = "Time (Seconds)",
    y = "Amplitude",
    colour = "Signal"
  ) +
  theme_minimal()

#-----
# Time Series Plot of Individual Input EEG Signals
#-----

ggplot(
  data_long,
  aes(
    x = Time,
    y = Amplitude
  )
) +

```

```

) +
  geom_line(color = "steelblue") +
  facet_wrap(
    ~Signal,
    ncol = 2,
    scales = "free_y"
  ) +
  labs(
    title = "Time Series Plot of Individual Input EEG Signals",
    x = "Time (Seconds)",
    y = "Amplitude"
  ) +
  theme_minimal()

#-----
# Output EEG Signal
#-----

data_y <- cbind(
  time,
  y
)

head(data_y, 10)

View(data_y)

#-----
# Time Series Plot of Output EEG Signal
#-----

ggplot(
  data_y,
  aes(
    x = Time,
    y = Y
  )
)

```

```

' +
  geom_line(
    color = "red",
    linewidth = 0.8
  ) +
  labs(
    title = "Time Series Plot of Output EEG Signal",
    x = "Time (Seconds)",
    y = "Amplitude"
  ) +
  theme_minimal()

#-----
# Combined Histogram of Input EEG Signals
#-----

library(ggplot2)
library(RColorBrewer)

ggplot(
  data_long,
  aes(
    x = Amplitude,
    fill = Signal
  )
) +
  geom_histogram(
    bins = 30,
    position = "identity",
    alpha = 0.5,
    color = "black"
  ) +
  scale_fill_brewer(
    palette = "Set1"
  ) +
  labs(
    title = "Combined Histogram of Input EEG Signals",

```

```

    title = "Combined Histogram of Input EEG Signals",
    x = "Amplitude",
    y = "Frequency",
    fill = "Input Signal"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(
      hjust = 0.5,
      face = "bold",
      size = 14
    ),
    axis.title = element_text(face = "bold"),
    legend.position = "top",
    legend.title = element_text(face = "bold")
  )

#-----
# Distribution Analysis
#-----

# Histogram of Input EEG Signals

ggplot(
  data_long,
  aes(x = Amplitude)
) +
  geom_histogram(
    bins = 30,
    fill = "steelblue",
    color = "black"
  ) +
  facet_wrap(
    ~Signal,
    ncol = 2,
    scales = "free"
  ) +
  labs(
    title = "Distribution of Input EEG Signals"
  )

```

```

  labs(
    title = "Distribution of Input EEG Signals",
    x = "Amplitude",
    y = "Frequency"
  ) +
  theme_minimal()

#-----
# Histogram of Output EEG Signal
#-----

ggplot(
  data_y,
  aes(x = Y)
) +
  geom_histogram(
    bins = 30,
    fill = "tomato",
    color = "black"
  ) +
  labs(
    title = "Distribution of Output EEG Signal",
    x = "Amplitude",
    y = "Frequency"
  ) +
  theme_minimal()

#-----
# Correlation Matrix
#-----
library(corrplot)
cor_data <- cbind(X, Y = y$Y)

cor_matrix <- cor(cor_data)

round(cor_matrix, 3)

corrplot(
  cor_matrix

```

```

corrplot(
  cor_matrix,
  method = "color",
  type = "upper",
  addCoef.col = "black",
  number.cex = 0.8,
  tl.col = "black",
  tl.srt = 45,
  diag = FALSE
)

#-----
# Scatter Plot Matrix
#-----

pairs(
  cor_data,
  main = "Scatter Plot Matrix of EEG Variables",
  pch = 19,
  col = "steelblue"
)

#-----
# Correlation of All X Inputs with Y
#-----
install.packages("tidyr") # Only run once if not installed
library(tidyr)
scatter_data <- data.frame(
  X1 = X$X1,
  X2 = X$X2,
  X3 = X$X3,
  X4 = X$X4,
  Y = y$Y
)

scatter_long <- pivot_longer(
  scatter_data,
  cols = c(X1, X2, X3, X4),
  names_to = "Signal",
  values_to = "Input"
)

```

```

ggplot(
  scatter_long,
  aes(
    x = Input,
    y = Y,
    colour = Signal
  )
) +
  geom_point(
    size = 2,
    alpha = 0.7
  ) +
  labs(
    title = "Correlation of X Inputs with Y Output",
    x = "Input Signal",
    y = "Output Signal (Y)",
    colour = "Signal"
  ) +
  theme_minimal()

```

```

#-----
# Individual Correlation Plots
#-----

```

```

ggplot(
  scatter_long,
  aes(
    x = Input,
    y = Y
  )
) +
  geom_point(
    color = "steelblue",
    size = 1.8,
    alpha = 0.7
  ) +
  facet_wrap(
    ~Signal,
    ncol = 2.
  )

```

```

    ~Signal,
    ncol = 2,
    scales = "free_x"
  ) +
  labs(
    title = "Correlation of Individual X Inputs with Y Output",
    x = "Input Signal",
    y = "Output Signal (Y)"
  ) +
  theme_minimal()

# Combine all variables into one dataset
EEG_Model1 <- data.frame(
  X1 = X$X1,
  X2 = X$X2,
  X3 = X$X3,
  X4 = X$X4,
  Y = y$Y
)

head(EEG_Model1)

# Model 1
model1 <- lm(
  Y ~ X4 +
    I(X1^2) +
    I(X1^3) +
    I(X2^4),
  data = EEG_Model1
)

summary(model1)

# Model 2
model2 <- lm(
  Y ~ X4 +
    I(X1^3) +
    I(X3^4),
  . . .
)

```

```

summary(model2)

# Model 3
model3 <- lm(
  Y ~ I(X3^3) +
    I(X2^4),
  data = EEG_Model
)

summary(model3)

# Model 4
model4 <- lm(
  Y ~ X2 +
    I(X1^3) +
    I(X3^4),
  data = EEG_Model
)

summary(model4)

# Model 5
model5 <- lm(
  Y ~ X4 +
    I(X1^2) +
    I(X1^3) +
    I(X3^4),
  data = EEG_Model
)

summary(model5)

# Calculate RSS for all models

rss1 <- sum(residuals(model1)^2)
rss2 <- sum(residuals(model2)^2)
rss3 <- sum(residuals(model3)^2)
rss4 <- sum(residuals(model4)^2)
rss5 <- sum(residuals(model5)^2)

```

```

rss_table <- data.frame(
  Model = c("Model 1", "Model 2", "Model 3", "Model 4", "Model 5"),
  RSS = c(rss1, rss2, rss3, rss4, rss5)
)

rss_table

# Calculate Log-Likelihood

loglik_table <- data.frame(
  Model = c("Model 1", "Model 2", "Model 3", "Model 4", "Model 5"),
  LogLikelihood = c(
    as.numeric(logLik(model1)),
    as.numeric(logLik(model2)),
    as.numeric(logLik(model3)),
    as.numeric(logLik(model4)),
    as.numeric(logLik(model5))
  )
)

loglik_table

# Create AIC and BIC comparison table

aic_bic_table <- data.frame(
  Model = c("Model 1", "Model 2", "Model 3", "Model 4", "Model 5"),
  AIC = c(
    AIC(model1),
    AIC(model2),
    AIC(model3),
    AIC(model4),
    AIC(model5)
  ),
  BIC = c(
    BIC(model1),
    BIC(model2),
    BIC(model3)
  )
)

```

```
#=====
# Residual Analysis - Model 1
#=====

residuals_model1 <- residuals(model1)

par(mfrow = c(1,2))

hist(
  residuals_model1,
  breaks = 20,
  col = "skyblue",
  border = "black",
  main = "Histogram (Model 1)",
  xlab = "Residuals"
)

qqnorm(
  residuals_model1,
  pch = 19,
  col = "darkblue",
  main = "Normal Q-Q Plot (Model 1)"
)

qqline(
  residuals_model1,
  col = "red",
  lwd = 2
)

par(mfrow = c(1,1))

#=====
# Residual Analysis - Model 2
#=====

residuals_model2 <- residuals(model2)

par(mfrow = c(1,2))
```

```
#=====
# Residual Analysis - Model 2
#=====

residuals_model2 <- residuals(model2)

par(mfrow = c(1,2))

hist(
  residuals_model2,
  breaks = 20,
  col = "lightgreen",
  border = "black",
  main = "Histogram (Model 2)",
  xlab = "Residuals"
)

qqnorm(
  residuals_model2,
  pch = 19,
  col = "darkgreen",
  main = "Normal Q-Q Plot (Model 2)"
)

qqline(
  residuals_model2,
  col = "red",
  lwd = 2
)

par(mfrow = c(1,1))
```

```
#=====
# Residual Analysis - Model 3
#=====

residuals_model3 <- residuals(model3)

par(mfrow = c(1,2))

hist(
  residuals_model3,
  breaks = 20,
  col = "gold",
  border = "black",
  main = "Histogram (Model 3)",
  xlab = "Residuals"
)

qqnorm(
  residuals_model3,
  pch = 19,
  col = "orange",
  main = "Normal Q-Q Plot (Model 3)"
)

qqline(
  residuals_model3,
  col = "red",
  lwd = 2
)

par(mfrow = c(1,1))

"
```

```
#=====
# Residual Analysis - Model 4
#=====

residuals_model4 <- residuals(model4)

par(mfrow = c(1,2))

hist(
  residuals_model4,
  breaks = 20,
  col = "plum",
  border = "black",
  main = "Histogram (Model 4)",
  xlab = "Residuals"
)

qqnorm(
  residuals_model4,
  pch = 19,
  col = "purple4",
  main = "Normal Q-Q Plot (Model 4)"
)

qqline(
  residuals_model4,
  col = "red",
  lwd = 2
)

par(mfrow = c(1,1))
..
```

```
#=====
# Residual Analysis - Model 5
#=====

residuals_model5 <- residuals(model5)

par(mfrow = c(1,2))

hist(
  residuals_model5,
  breaks = 20,
  col = "tomato",
  border = "black",
  main = "Histogram (Model 5)",
  xlab = "Residuals"
)

qqnorm(
  residuals_model5,
  pch = 19,
  col = "brown",
  main = "Normal Q-Q Plot (Model 5)"
)

qqline(
  residuals_model5,
  col = "blue",
  lwd = 2
)

par(mfrow = c(1,1))
```

```

#-----
# Task 2.6: Model Selection
#-----

# Create Model Comparison Table

model_selection <- data.frame(
  Model = c("Model 1", "Model 2", "Model 3", "Model 4", "Model 5"),
  RSS = c(
    sum(residuals(model1)^2),
    sum(residuals(model2)^2),
    sum(residuals(model3)^2),
    sum(residuals(model4)^2),
    sum(residuals(model5)^2)
  ),
  LogLikelihood = c(
    as.numeric(logLik(model1)),
    as.numeric(logLik(model2)),
    as.numeric(logLik(model3)),
    as.numeric(logLik(model4)),
    as.numeric(logLik(model5))
  ),
  AIC = c(
    AIC(model1),
    AIC(model2),
    AIC(model3),
    AIC(model4),
    AIC(model5)
  ),
  BIC = c(
    BIC(model1),
    BIC(model2),
    BIC(model3),
    BIC(model4),
    BIC(model5)
  )
)

```

```

# Round values for presentation
model_selection[,2:5] <- round(model_selection[,2:5], 3)

# Display the comparison table
model_selection

#-----
# Select Best Model
#-----

best_model <- model_selection[
  which.min(model_selection$AIC),
]

best_model

#=====
# Task 2.7: Model Evaluation
# Selected Model: Model 2
#=====

# Predicted Values
predicted <- predict(model2)

# Actual Values
actual <- EEG_Model$Y

#-----
# Performance Measures
#-----

RMSE <- sqrt(mean((actual - predicted)^2))

MAE <- mean(abs(actual - predicted))

R2 <- summary(model2)$r.squared

```

```
Adj_R2 <- summary(model2)$adj.r.squared
```

```
evaluation_table <- data.frame(  
  Measure = c(  
    "RMSE",  
    "MAE",  
    "R-Squared",  
    "Adjusted R-Squared"  
  ),  
  Value = round(  
    c(  
      RMSE,  
      MAE,  
      R2,  
      Adj_R2  
    ),  
    4  
  )  
)
```

```
evaluation_table
```

```
#-----  
# Actual vs Predicted Plot  
#-----
```

```
plot(  
  actual,  
  predicted,  
  pch = 19,  
  col = "steelblue",  
  main = "Actual vs Predicted Values",  
  xlab = "Actual Output",  
  ylab = "Predicted Output"  
)
```

```
abline(  
  ~
```

```

#-----
# Residuals vs Fitted Plot
#-----

plot(
  fitted(model2),
  residuals(model2),
  pch = 19,
  col = "darkgreen",
  main = "Residuals vs Fitted Values",
  xlab = "Fitted Values",
  ylab = "Residuals"
)

abline(
  h = 0,
  col = "red",
  lwd = 2,
  lty = 2)

#=====
# Task 2.7
# Step 1: Split Dataset into Training and Testing Sets
#=====
if (nrow(EEG_Model) == 201) {
  EEG_Model <- EEG_Model[-1, ]
}

# Set seed for reproducibility
set.seed(123)

# Total number of observations
n <- nrow(EEG_Model)

# Select 70% of observations for training
train_index <- sample(

```

```

# Function to Display Dataset Size
#-----

display_observations <- function(train_data, test_data) {

  cat(
    "Training set =",
    nrow(train_data),
    "observations\n"
  )

  cat(
    "Testing set =",
    nrow(test_data),
    "observations\n"
  )

}

# Call the function
display_observations(train_data, test_data)


#=====
# Task 2.7
# Step 2: Train the Selected Model (Model 2)
#=====

# Fit Model 2 using the training dataset
model2_train <- lm(
  Y ~ X4 +
    I(X1^3) +
    I(X3^4),
  data = train_data
)

# Display model summary
summary(model2_train)

```

```

#=====
# Task 2.7
# Step 4: Plot Predictions with 95% Prediction Intervals
#=====

plot(
  test_data$Y,
  prediction_results[, "fit"],
  pch = 19,
  col = "blue",
  xlab = "Observed Output",
  ylab = "Predicted Output",
  main = "Observed vs Predicted Output (95% Prediction Intervals)"
)

# Identity line
abline(
  0,
  1,
  col = "red",
  lwd = 2
)

# Add 95% prediction interval error bars
arrows(
  x0 = test_data$Y,
  y0 = prediction_results[, "lwr"],
  x1 = test_data$Y,
  y1 = prediction_results[, "upr"],
  angle = 90,
  code = 3,
  length = 0.05,
  col = "darkgreen"
)

```

```
# Final Prediction Plot with 95% Prediction Intervals
```

```
#=====
```

```
prediction_df <- data.frame(  
  Observation = 1:nrow(test_data),  
  Actual = test_data$Y,  
  Predicted = prediction_results[, "fit"],  
  Lower = prediction_results[, "lwr"],  
  Upper = prediction_results[, "upr"]  
)  
  
plot(  
  prediction_df$Observation,  
  prediction_df$Predicted,  
  type = "b",  
  pch = 19,  
  col = "blue",  
  ylim = range(  
    prediction_df$Lower,  
    prediction_df$Upper  
  ),  
  xlab = "Testing Observation",  
  ylab = "EEG Output",  
  main = "Prediction with 95% Prediction Intervals"  
)  
  
points(  
  prediction_df$Observation,  
  prediction_df$Actual,  
  pch = 17,  
  col = "red"  
)  
  
arrows(  
  prediction_df$Observation,  
  prediction_df$Lower,  
  prediction_df$Observation,  
  prediction_df$Upper,  
  angle = 90  
)
```

```
legend(  
  "topright",  
  legend = c(  
    "Predicted",  
    "Actual",  
    "95% Prediction Interval"  
  ),  
  col = c(  
    "blue",  
    "red",  
    "darkgreen"  
  ),  
  pch = c(  
    19,  
    17,  
    NA  
  ),  
  lty = c(  
    1,  
    NA,  
    1  
  ),  
  lwd = c(  
    1,  
    NA,  
    2  
  )  
)
```

```

#=====
# Task 3
# Step 1 : Extract Estimated Parameters
#=====

coef(model2)

summary(model2)$coefficients

#=====
# Task 3
# Step 2 : Define Uniform Prior Distributions
#=====

# Estimated coefficients from Model 2
beta0_hat <- coef(model2)[1]
beta1_hat <- coef(model2)[2]

# Uniform prior limits ( $\pm 50\%$ )
beta0_lower <- beta0_hat * 0.5
beta0_upper <- beta0_hat * 1.5

beta1_lower <- beta1_hat * 0.5
beta1_upper <- beta1_hat * 1.5

# Display prior ranges
prior_table <- data.frame(
  Parameter = c("Intercept", "X4"),
  Lower = c(beta0_lower, beta1_lower),
  Upper = c(beta0_upper, beta1_upper)
)

prior_table

```

```

#=====
# Task 3
# Step 3 : Sample from Uniform Priors
#=====

set.seed(123)

N <- 10000

beta0_prior <- runif(
  N,
  min = beta0_lower,
  max = beta0_upper
)

beta1_prior <- runif(
  N,
  min = beta1_lower,
  max = beta1_upper
)

# Display first few samples
head(beta0_prior)
head(beta1_prior)

#=====
# Task 3
# Step 4 : Rejection ABC
#=====

# Fixed parameters from Model 2
beta2 <- coef(model2)["I(X1^3)"]
beta3 <- coef(model2)["I(X3^4)"]

# Observed output
y_obs <- EEG_Model$Y

# Number of prior samples
N <- length(beta0_prior)

```

```

# Loop through each sampled parameter pair
for(i in 1:N){

  # Model prediction
  y_sim <-
    beta0_prior[i] +
    beta1_prior[i] * EEG_Model$X4 +
    beta2 * (EEG_Model$X1^3) +
    beta3 * (EEG_Model$X3^4)

  # Euclidean distance
  distance[i] <-
    sqrt(sum((y_obs - y_sim)^2))

}
head(distance)

#=====
# Task 3
# Step 5 : Accept Posterior Samples
#=====

# Acceptance threshold (5%)
epsilon <- quantile(
  distance,
  0.05
)

accepted <- distance <= epsilon

# Posterior samples
posterior_beta0 <- beta0_prior[accepted]
posterior_beta1 <- beta1_prior[accepted]

cat(
  "Accepted Samples =",
  length(posterior_beta0),
  "\n"
)

```

```

# task 3
# Marginal Posterior Distribution of  $\beta_0$ 
#=====

hist(
  posterior_beta0,
  breaks = 30,
  probability = TRUE,
  col = "skyblue",
  border = "black",
  main = expression("Marginal Posterior of " * beta[0]),
  xlab = expression(beta[0])
)

lines(
  density(posterior_beta0),
  col = "red",
  lwd = 2
)

abline(
  v = beta0_hat,
  col = "blue",
  lwd = 2,
  lty = 2
)

legend(
  "topright",
  legend = c("Posterior Density", "Least Squares Estimate"),
  col = c("red", "blue"),
  lwd = 2,
  lty = c(1,2)
)

#=====
# Marginal Posterior Distribution of  $\beta_1$ 
#=====

```

```

#=====
# Marginal Posterior Distribution of  $\beta_1$ 
#=====

hist(
  posterior_beta1,
  breaks = 30,
  probability = TRUE,
  col = "lightgreen",
  border = "black",
  main = expression("Marginal Posterior of " * beta[1]),
  xlab = expression(beta[1])
)

lines(
  density(posterior_beta1),
  col = "red",
  lwd = 2
)

abline(
  v = beta1_hat,
  col = "blue",
  lwd = 2,
  lty = 2
)

legend(
  "topright",
  legend = c("Posterior Density", "Least Squares Estimate"),
  col = c("red", "blue"),
  lwd = 2,
  lty = c(1,2)
)

```

```
# Joint Posterior Distribution
#=====

plot(
  posterior_beta0,
  posterior_beta1,
  pch = 19,
  col = rgb(0,0,1,0.4),
  xlab = expression(beta[0]),
  ylab = expression(beta[1]),
  main = "Joint Posterior Distribution"
)

points(
  beta0_hat,
  beta1_hat,
  pch = 19,
  col = "red",
  cex = 2
)

legend(
  "topright",
  legend = c(
    "Posterior Samples",
    "Least Squares Estimate"
  ),
  col = c(
    rgb(0,0,1,0.4),
    "red"
  ),
  pch = 19
)
```

```

#=====
# Posterior Summary
#=====
posterior_summary <- data.frame(
  Parameter = c("Intercept ( $\beta_0$ )", "x4 ( $\beta_1$ )"),
  Mean = c(
    mean(posterior_beta0),
    mean(posterior_beta1)
  ),
  Median = c(
    median(posterior_beta0),
    median(posterior_beta1)
  ),
  SD = c(
    sd(posterior_beta0),
    sd(posterior_beta1)
  )
)
|
# Round only the numeric columns
posterior_summary[, 2:4] <- round(posterior_summary[, 2:4], 4)

posterior_summary

```